

# Modèles en physiologie cardiaque

*(Models in cardiac physiology)*

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UE MU4BI006

Diversité des modèles d'études en physiologie

# Avant de commencer

## Pré-requis

- ✓ Connaissances (niveau requis: L3) en physiologie cardiovasculaire:
  - Anatomie cardiaque
  - Cycle cardiaque
  - Electrophysiologie
  - Les cellules cardiaques
  - Maladies cardiaques

## Liens avec les autres enseignements de l'UE

- ✓ Cours « Les modèles animaux KO-KI » (A. L'Honoré)
- ✓ Cours « Culture des cellules animales » (V. Béréziat)
- ✓ Cours « Organoïdes et cellules souches » (G. Orioux)
- ✓ TD4 « Modèles de Régénération Cardiaque »

# Objectifs

- Avoir une connaissance globale des modèles d'études les plus courants en physiologie et physiopathologie cardiovasculaires
- Comprendre comment choisir un modèle expérimental
- Pouvoir établir un argumentaire précis qui permet de justifier l'utilisation d'un modèle expérimental
- Comprendre les principes généraux des méthodes expérimentales
- Réussir à analyser et interpréter des résultats scientifiques

# Plan général du cours

## **I. Introduction**

- Choisir un modèle
- Diversité et caractéristiques des modèles en physio cardiaques

## **II. Les modèles animaux (*in vivo*)**

- un peu d'Histoire
- Anatomie comparée
- Différents modèles animaux (poisson, poulet et mammifères)
- Les modèles murins
- Modèles ischémiques et non-ischémiques

## **III. Modèles *ex vivo***

- Le cœur isolés perfusé selon la méthode de Langendorff

## **Les modèles cellulaires (*in vitro*)**

- Les cellules cardiaques et les différents modèles cellulaires
- Les cellules souches pluripotentes
- Différentiation cardiaque
- Maturité des hiPSC-CMs
- Générer un modèle génétique

## **V. Modèles 3D**

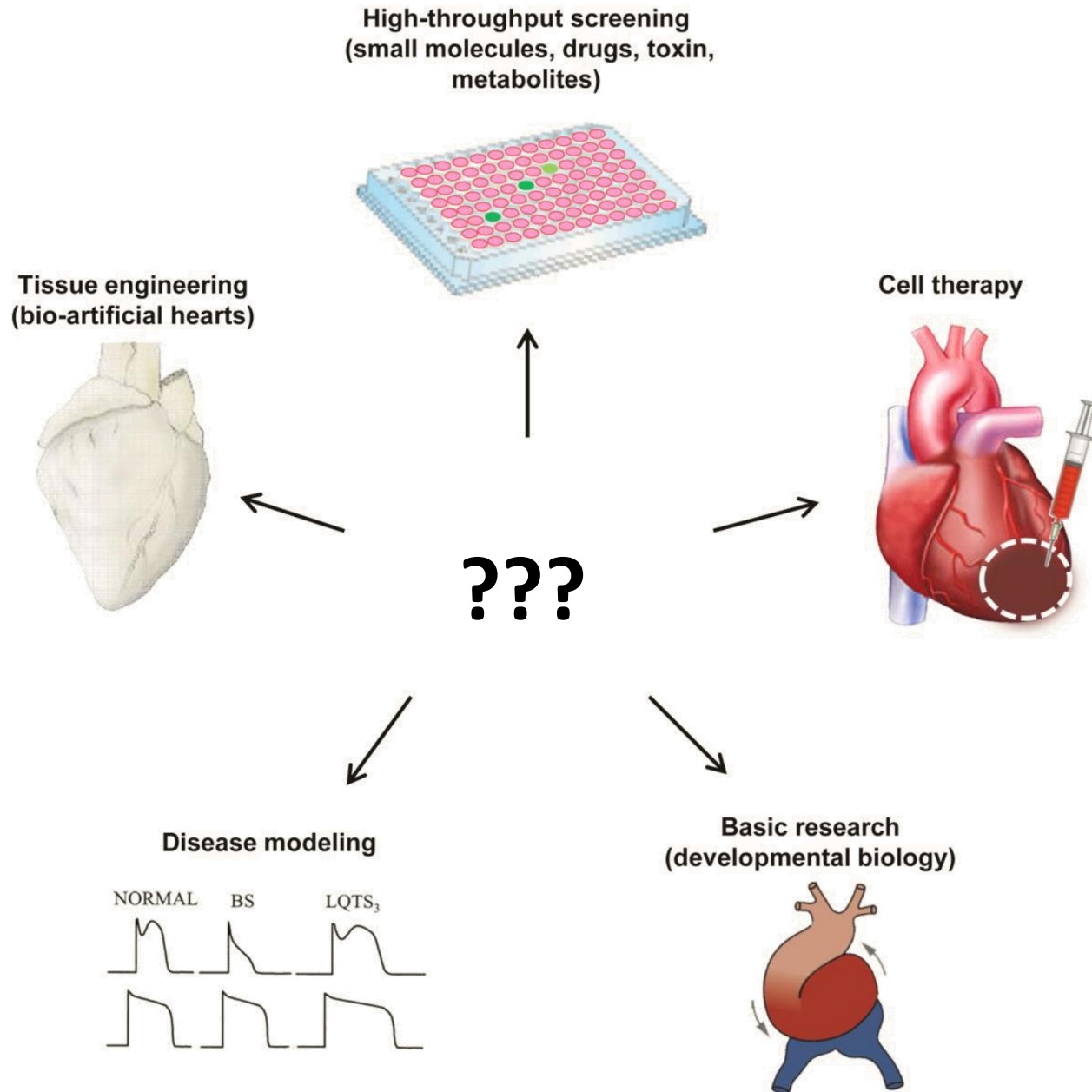
- Les différents modèles tissulaires
- Le modèle sphéroïde

## **VI. Se préparer à l'examen**

- Sujet d'examen et corrigé

# **I. Introduction**

# What are the biological questions in cardiac physiology?

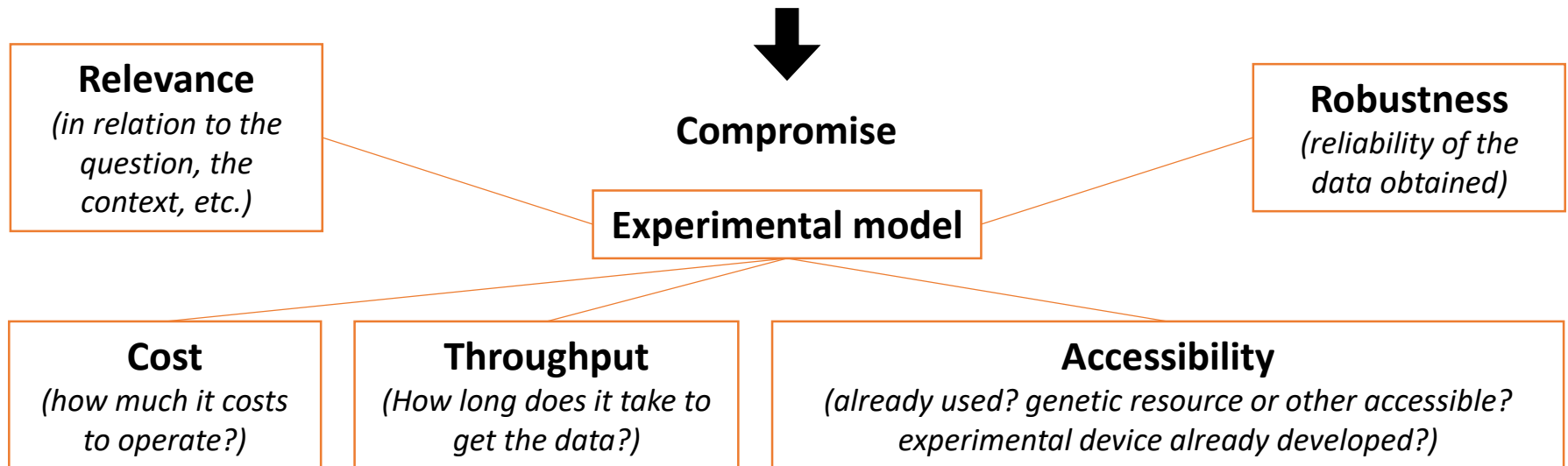


# Choose an experimental model

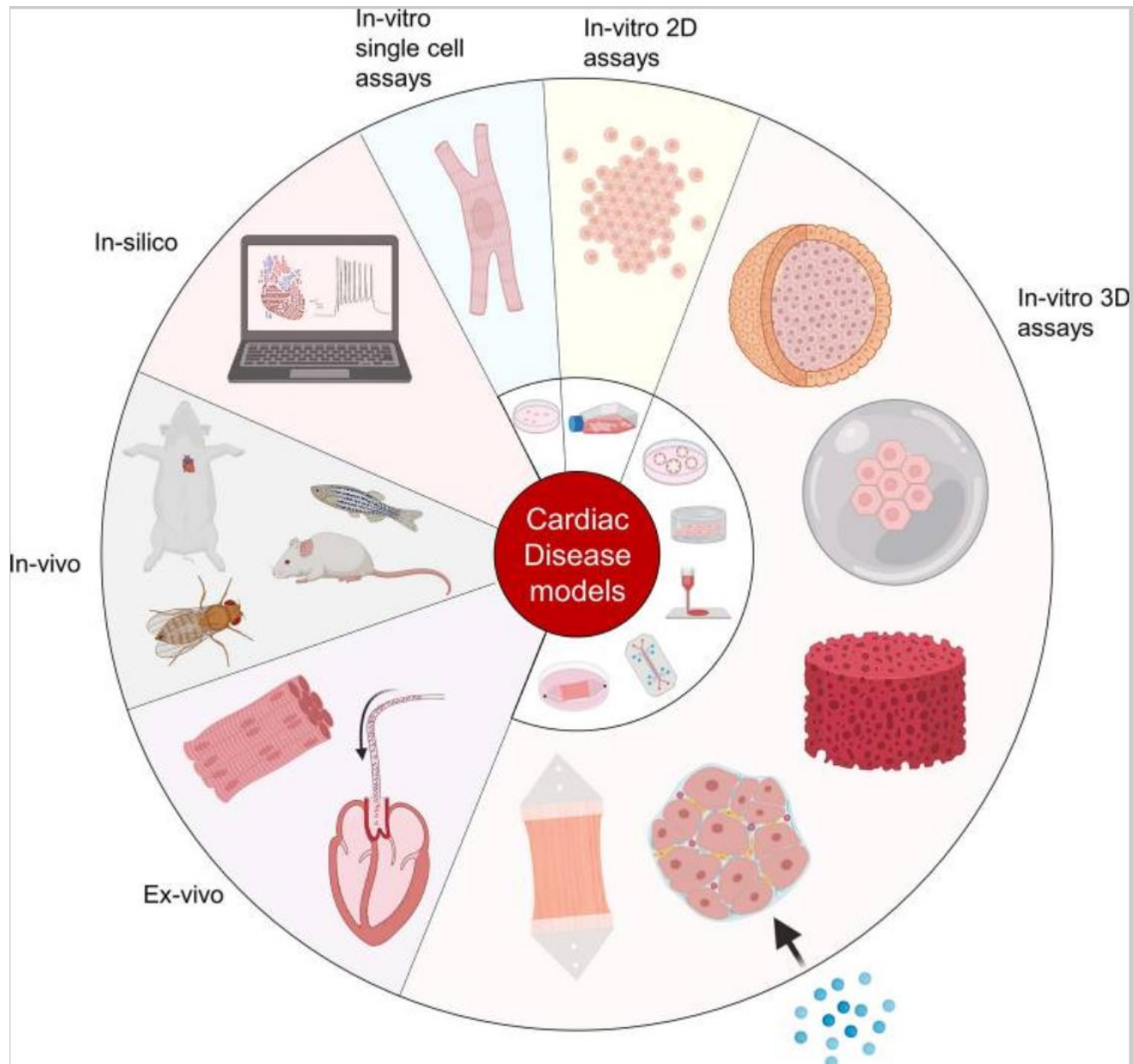
→ What is the scientific question at the origin of the experiments?

→ What are the resources available:

- Financial resources
- Time
- Human and technical resources



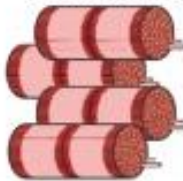
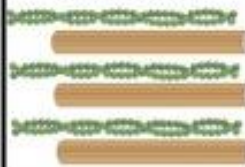
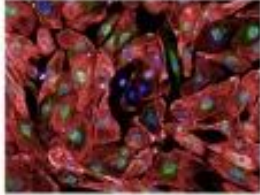
# Diversity of models for cardiac physiology



# Main characteristics of experimental models

|          | In vitro                                                                                                                                                                         | Ex vivo                                                                                                                                                                                                      | In vivo                                                                                                                                                                                                       |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Features | <ul style="list-style-type: none"><li>• Relatively low cost</li><li>• Cardiomyocyte-specific assessments</li><li>• Minimize animal sacrifice</li><li>• High-throughput</li></ul> | <ul style="list-style-type: none"><li>• Controlled experimental setting</li><li>• Limited neurological and hormonal influences</li><li>• Able to assess pump function</li><li>• Acute studies only</li></ul> | <ul style="list-style-type: none"><li>• Incorporates biological complexity</li><li>• Clinically relevant disease models</li><li>• High cost and low throughput</li><li>• Long-term studies possible</li></ul> |

# Advantages et disadvantages of models

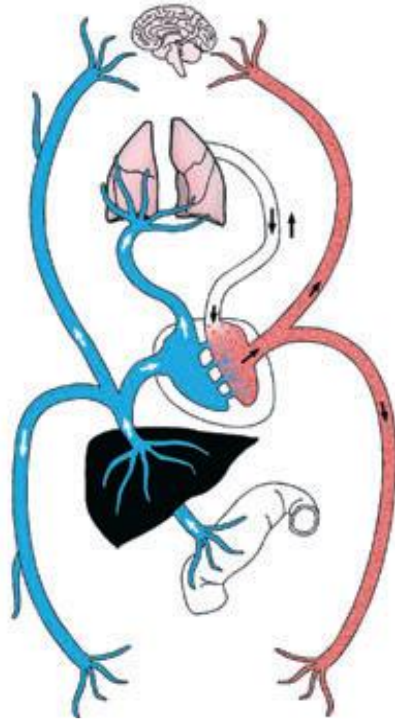
|                      | Intact heart muscle strips                                                                                    | Isolated cardiomyocytes                                                                            | Myofibrils from tissue/cells                                                              | Purified actomyosin                                                                                             | Animal models                                                                                        | hPSC-CMs                                                                                                                 |
|----------------------|---------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
| <b>Disease model</b> |                              |                   |          |                              |                   |                                       |
| <b>Advantages</b>    | Physiological relevance<br>Pharmacological predictivity                                                       | Amenable to measurements of subcellular functions                                                  | Direct access to myofilament function<br>Simple logistics (allows sample freezing)        | Higher sample availability (heterologous systems)<br>Direct assessment of actin–myosin interactions             | Whole organ/organism studies<br>Genetically tractable<br>Easily available cell and tissue samples    | Unlimited source of cardiomyocytes<br>Recaptures patient genotype<br>Gene editing enables isogenic controls              |
| <b>Disadvantages</b> | Demanding logistics<br>Limited sample availability                                                            | Limited sample availability<br>Lack of proliferative ability in culture<br>Lack of 3D architecture | Preparation artefacts<br>Limited sample availability                                      | Over-simplistic (nonrepresentative of whole sarcomere)<br>Preparation artefacts (causing contradictory results) | Species differences relative to human<br>Time-consuming genetic manipulation                         | Immaturity relative to human adult cardiomyocytes<br>Lack of multicellular cardiac composition and neurohormonal control |
| <b>Main readouts</b> | Histology: fibrosis, myocyte enlargement, and disarray<br>Pharmacological responses (e.g., $\beta$ -blockers) | Electrophysiology<br>Single-cell transcriptomics                                                   | Calcium sensitivity<br>Isometric tension<br>ATP consumption<br>Energy cost of contraction | Kinetics of contraction<br>ATPase activity<br>Surrogate of force generation                                     | Histology: fibrosis, disarray<br>Contraction parameters and calcium sensitivity (integrated in EHTs) | High-throughput molecular and functional phenotyping<br>Genetic causation (isogenic controls)                            |

Trends in Molecular Medicine

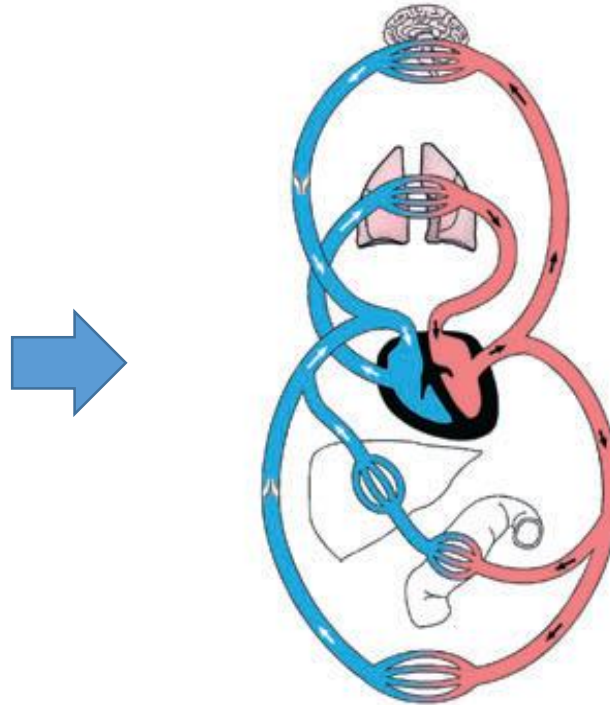
## II. Animal models (*in vivo*)

# A short story of the blood circulation

**Claude Galien, -160**



**William Harvey, 1628**

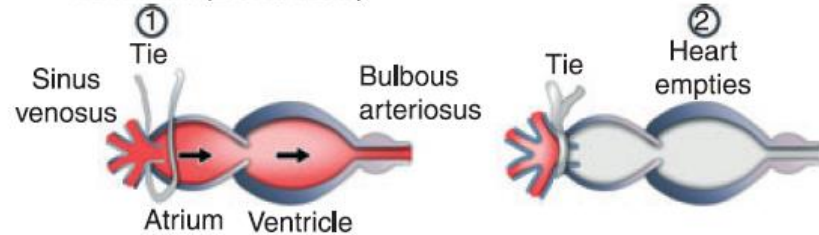


## **Galenic conception of blood circulation:**

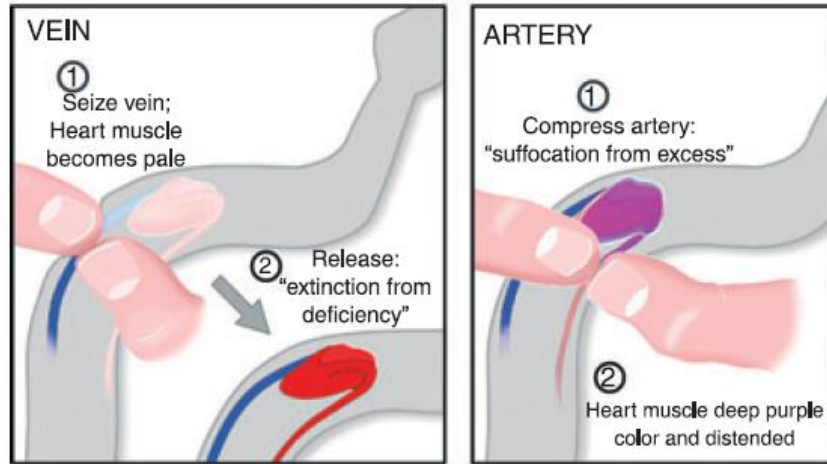
- Liver = main role of distributing organ
- Veins arise from the liver, arteries from the heart
- Venous blood is transformed into arterial blood in the left ventricle
- Blood is formed in the liver which transforms food coming from the digestive tract
- Venous blood is sent to the lungs and frees itself of its impurities and also passes into the left ventricle through pores

# The models used by Harvey

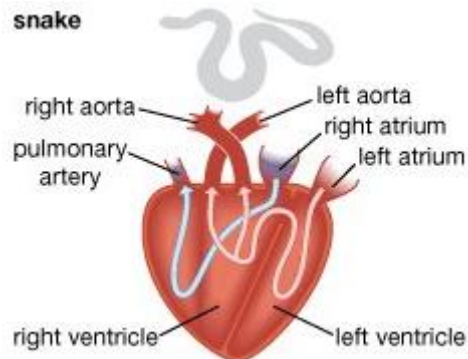
**A** Fish heart (2 chambers)



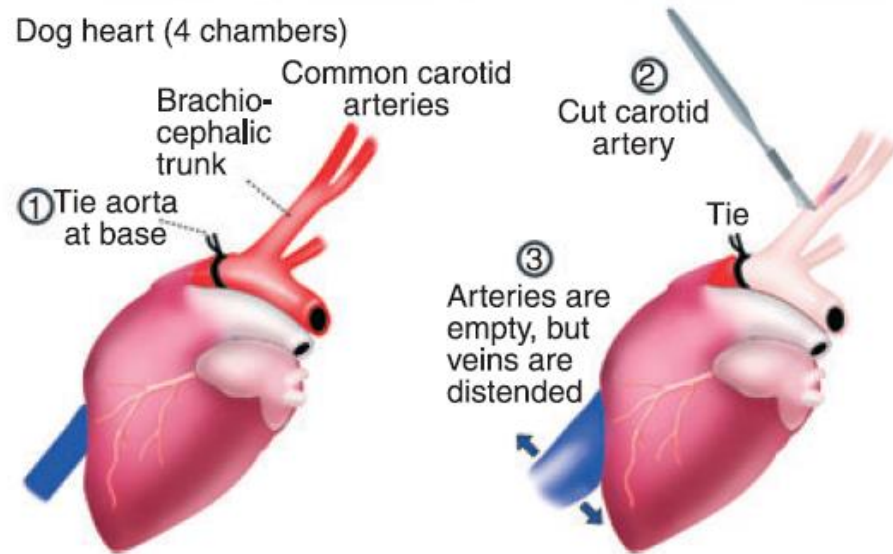
**B** Snake heart (3 chambers)



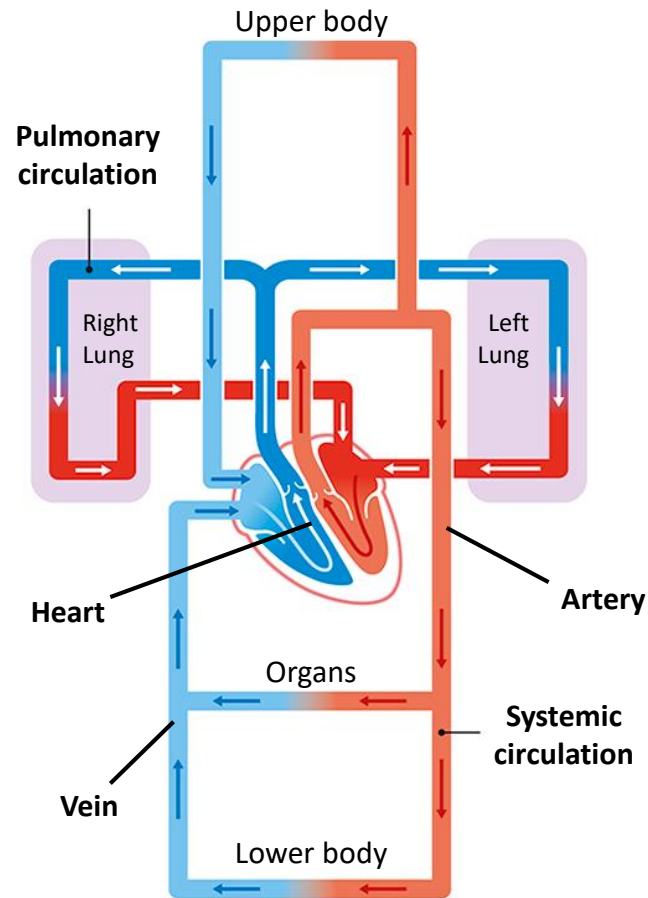
**snake**



**C** Dog heart (4 chambers)



# The cardiovascular system



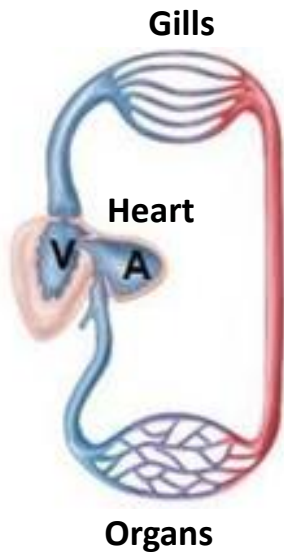
## Role of the cardiovascular system:

- Ensure tissue perfusion allowing cellular oxygenation, but also the transport of carbon dioxide, metabolic, hormonal and immunological compounds essential to cells and various organic functions.

# Comparative anatomy

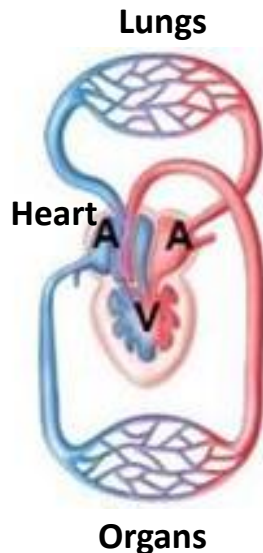
## Fish

1 circuit  
2 chambers



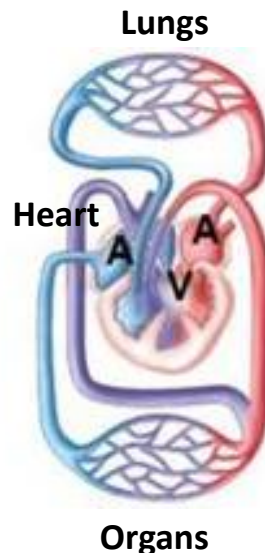
## Frog

2 circuits  
3 chambers



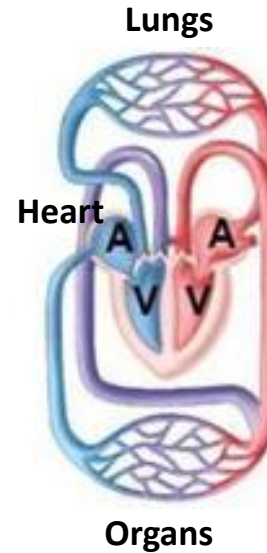
## Turtle, lizard

2 circuits  
5 chambers



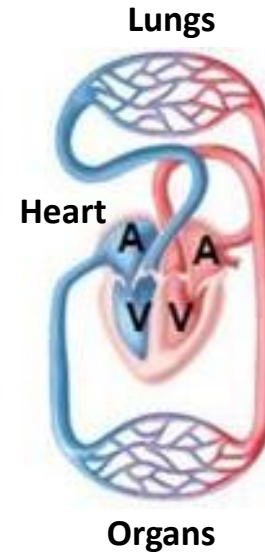
## Crocodiles

2 circuits  
4 chambers



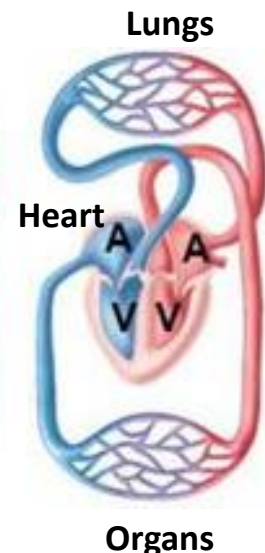
## Bird

2 circuits  
4 chambers

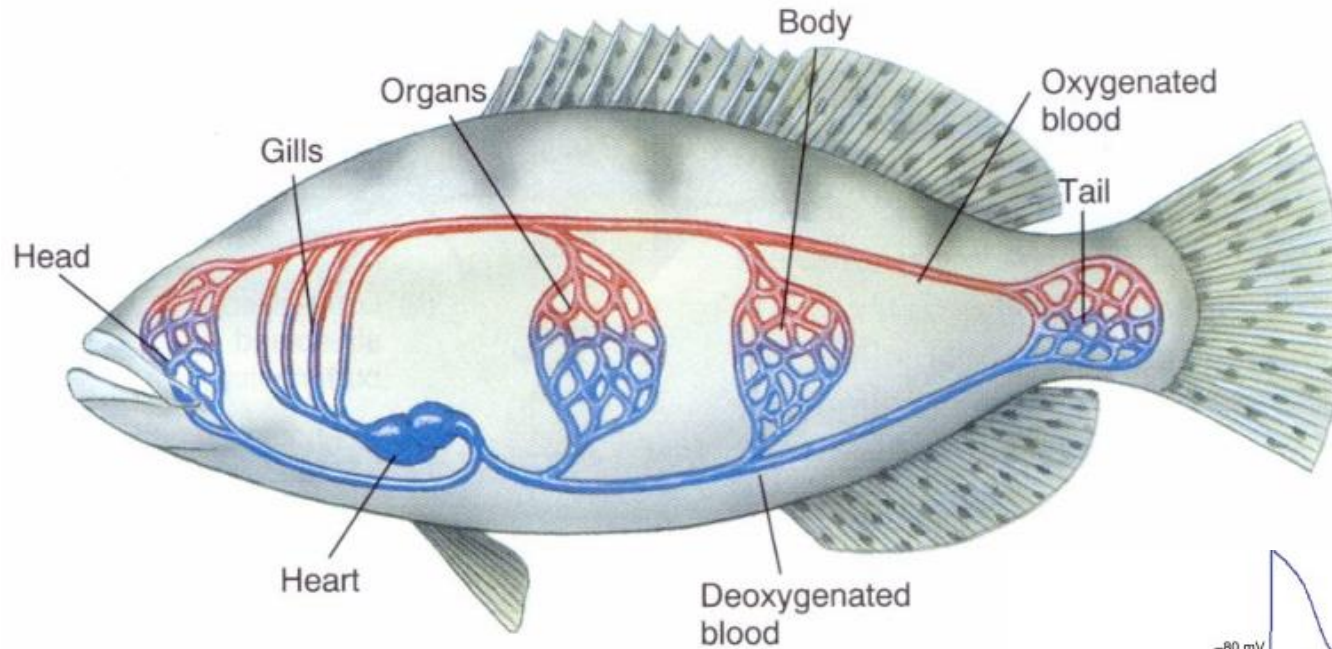


## Mammal

2 circuits  
4 chambers

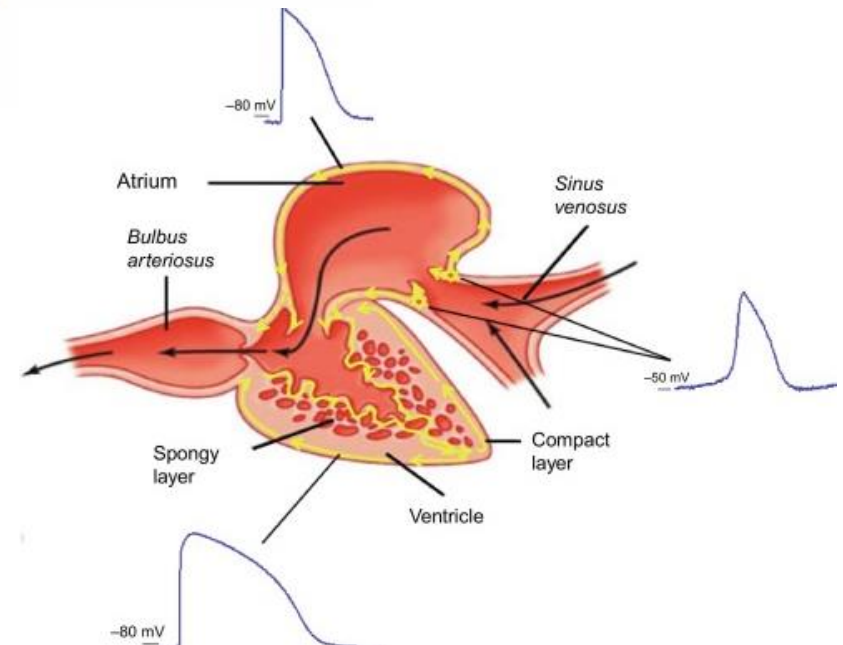


# The cardiovascular system in fish

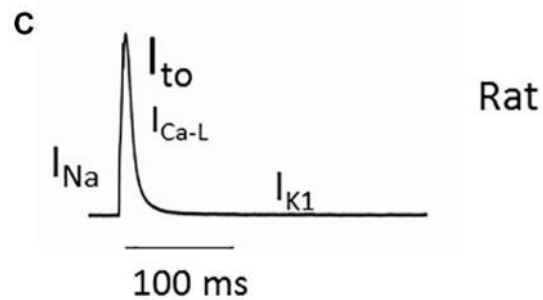
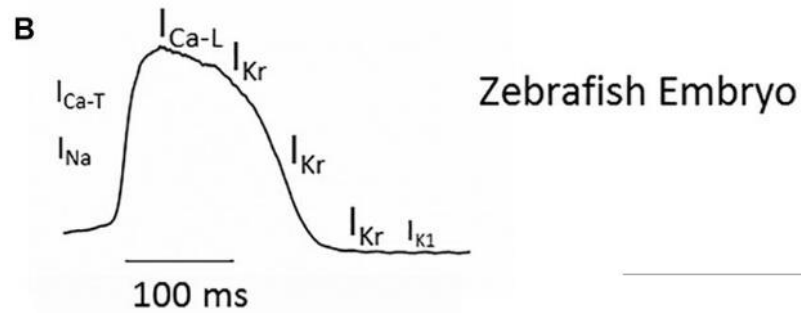
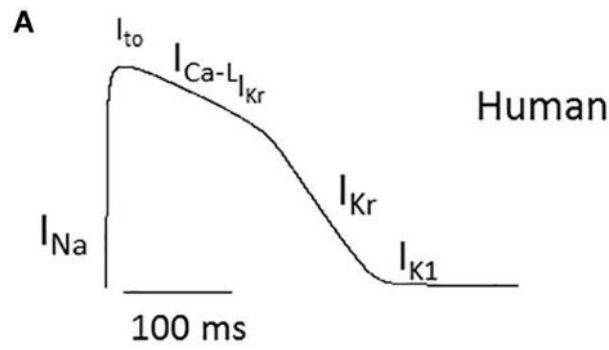


Using the fish model (mainly zebrafish):

- ➔ Heart development
- ➔ Cardiac regeneration



# Cardiac electrophysiology in zebrafish



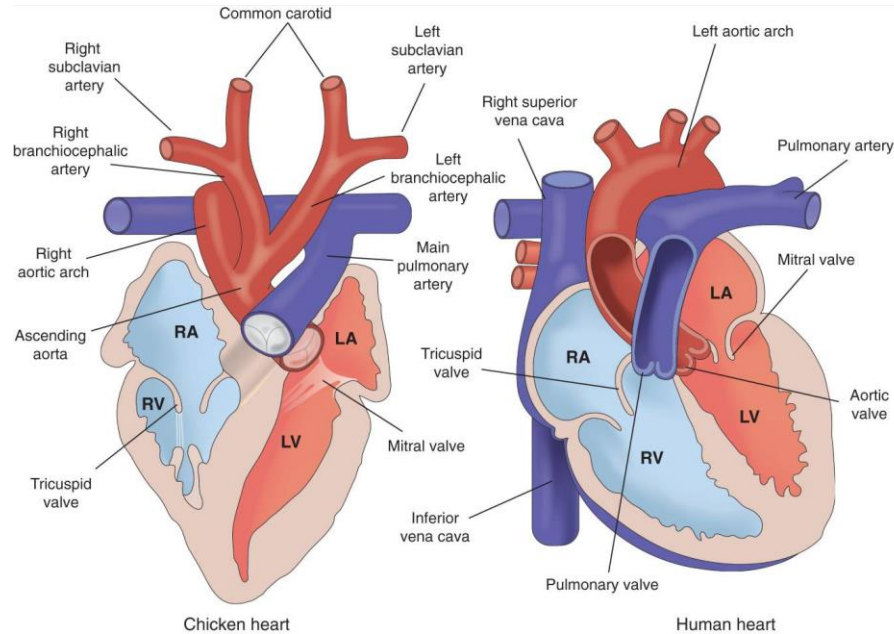
**Electrophysiology very similar to humans:**

➔ Cardiotoxicity model

|            | Adult Human      |                            | Zebrafish Embryo |                            | Adult Zebrafish |                            |
|------------|------------------|----------------------------|------------------|----------------------------|-----------------|----------------------------|
|            | Channel          | Accessory subunit          | Channel          | Accessory subunit          | Channel         | Accessory subunit          |
| $I_{Na}$   | Nav1.5           | Nav $\beta$                | Nav1.5           | ??                         | Nav1.5          | Nav $\beta$                |
| $I_{Ca-L}$ | Cav1.2           | $\beta$ ; $\alpha 2\delta$ | Cav1.2           | $\beta$ ; $\alpha 2\delta$ | Cav1.2          | $\beta$ ; $\alpha 2\delta$ |
| $I_{Ca-T}$ | –                | –                          | Cav3.2           | –                          | Cav3.2          | –                          |
| $I_{to}$   | Kv4.3            | MIRP1                      | –                | –                          | –               | –                          |
| $I_{Kr}$   | hERG1            |                            | hERG?            |                            | hERG2           |                            |
| $I_{Ks}$   | Kv71.            | MinK                       | –                | –                          | Kv7.1           | –                          |
| $I_{K1}$   | Kir2.1; 2.2; 2.3 | –                          | Kir2.4           | –                          | Kir2.4          | –                          |
| $I_K$      | –                | –                          | –                | –                          | –               | –                          |

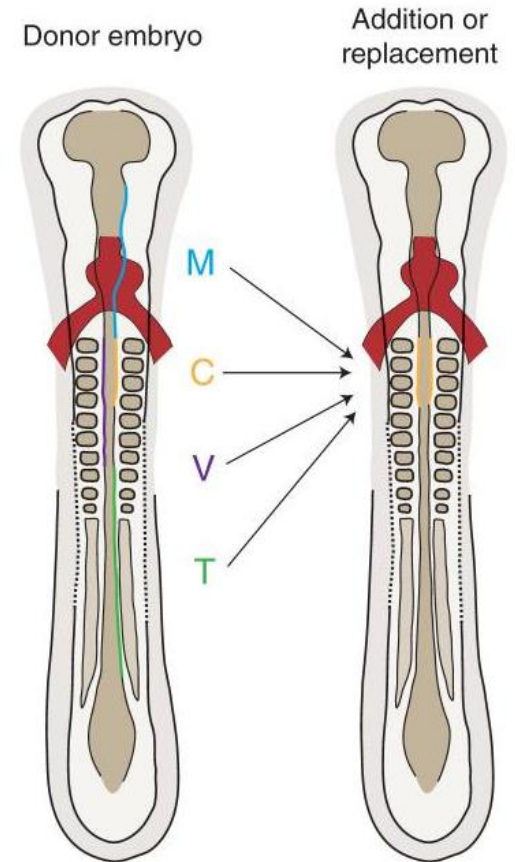
# Heart development: the example of the chicken

## Comparison of cardiac anatomy



## Fate mapping

(cf Le Douarin 1973))

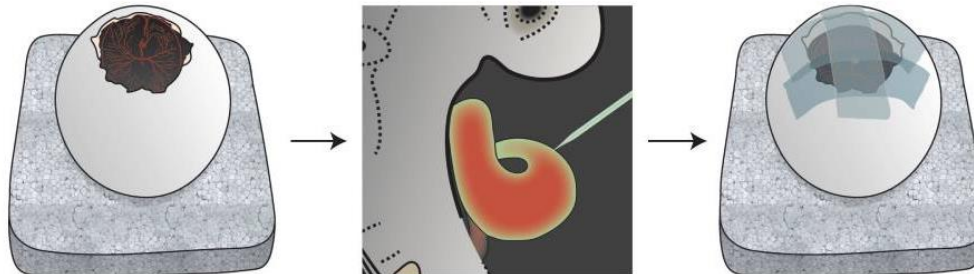


## Cardiac injection

Crack egg and inject ink

Access heart and inject compound

Lower embryo and seal egg



# Paramètres fonctionnels des modèles mammifères

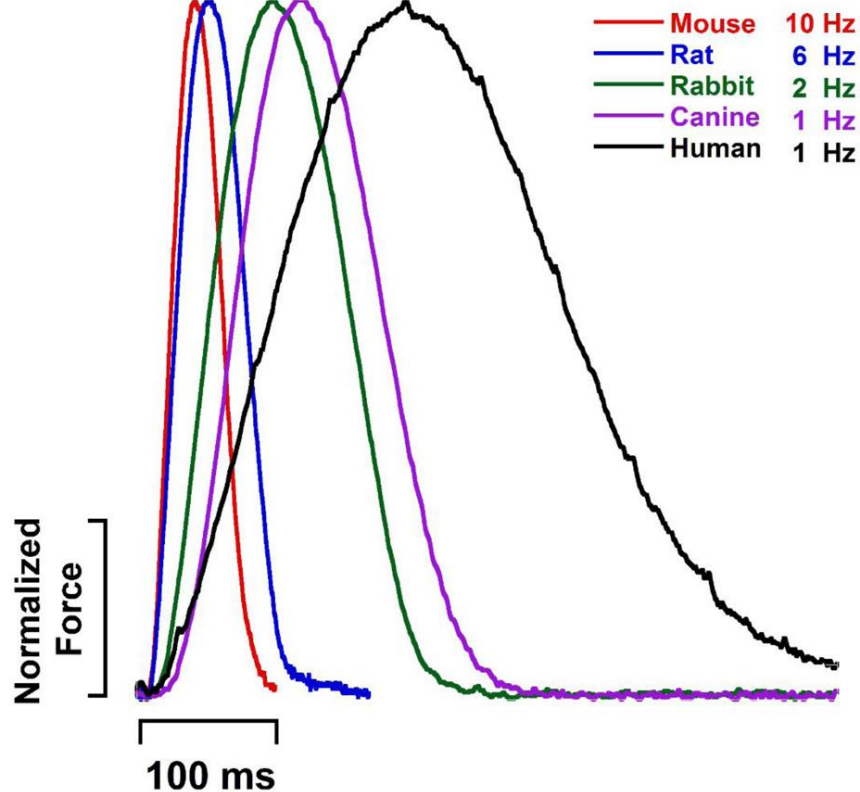


| Species | Body weight (kg) | Cardiac rythm (bpm) | Systolic pressure (mmHg) | Diastolic pressure (mmHg) |
|---------|------------------|---------------------|--------------------------|---------------------------|
| Mouse   | 0,02-0,063       | 310-840             | 113-160                  | 81-110                    |
| Rat     | 0,225-0,52       | 250-493             | 84-184                   | 58-145                    |
| Rabbit  | 1-6              | 130-300             | 90-130                   | 60-90                     |
| Dog     | 7-16             | 70-160              | 95-136                   | 43-66                     |
| Sheep   | 20-160           | 60-120              | 91-116                   | 102                       |
| Pig     | 200-300          | 50-116              | 135-150                  | -                         |
| Human   | 50-86            | 72                  | 120                      | 80                        |

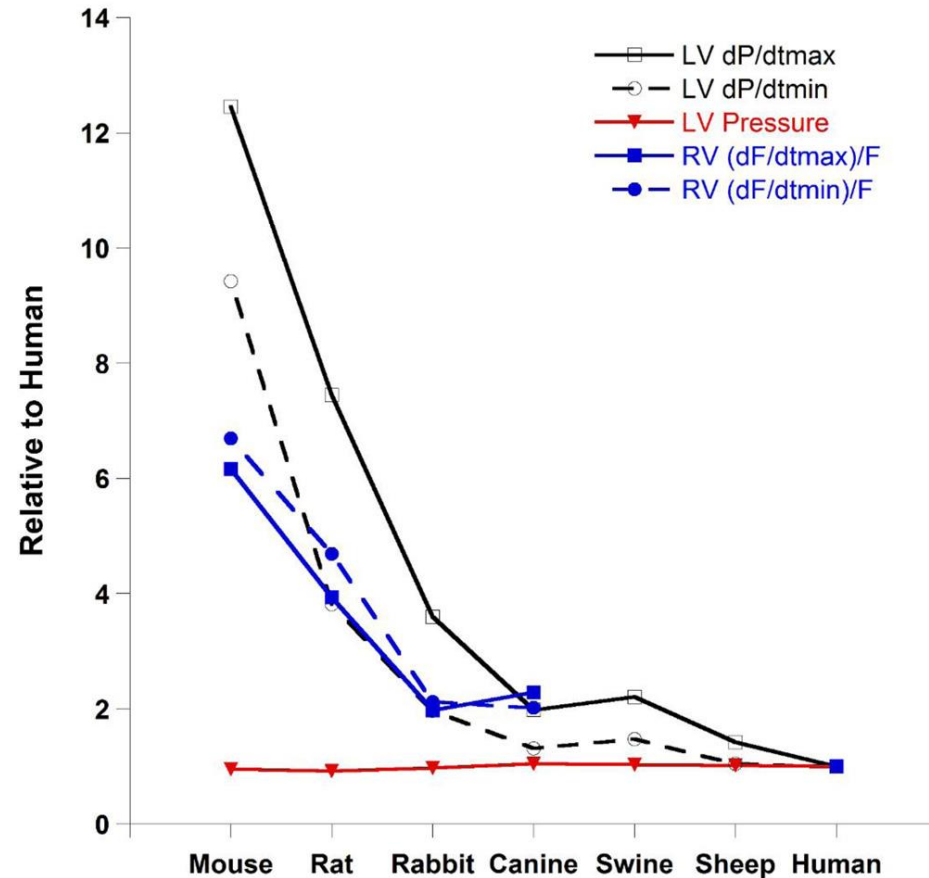


# Cardiac contraction in mammalian models

Cardiac ventricular muscle contraction of various species near their respective *in vivo* heart rates



Contraction and Relaxation Kinetics in Small vs. Large Animal Models



# Murine models of cardiac physiology

| Paramètres                                                                        | Souris  | Rat    | Humain | Ratio S/H | Ratio R/H |
|-----------------------------------------------------------------------------------|---------|--------|--------|-----------|-----------|
| Force (mN/mm <sup>2</sup> )                                                       | 40      | 27,8   | 16,7   | 2,4       | 1,66      |
| Time to peak (ms)                                                                 | 36,3    | 54,2   | 235    | 0,15      | 0,23      |
| Relaxation time 50% (ms)                                                          | 20,4    | 29,4   | 153    | 0,13      | 0,19      |
| Maximal velocity of contraction (dF/dt <sub>max</sub> , mN/mm <sup>2</sup> /s)    | 1899,5  | 842,9  | 129    | 14,72     | 6,53      |
| Maximal velocity of relaxation (dF/dt <sub>min</sub> , mN/mm <sup>2</sup> /s)     | -1579,8 | -768,7 | -98,3  | 16,07     | 7,82      |
| Contraction kinetic ((dF/dt <sub>max</sub> )/F, s <sup>-1</sup> )                 | 47,5    | 30,3   | 7,7    | 6,17      | 3,94      |
| Relaxation kinetic ((dF/dt <sub>min</sub> )/F, s <sup>-1</sup> )                  | -39,5   | -27,7  | -5,9   | 6,69      | 4,69      |
| Left ventricular systolic pressure (mmHg)                                         | 110     | 106    | 115    | 0,96      | 0,92      |
| Left ventricular isovolumetric relaxation time constant (τ, ms)                   | 4,4     | 6,3    | 43     | 0,10      | 0,15      |
| Left ventricular maximal rate of pressure rise (dP/dt <sub>max</sub> , mmHg/s)    | 18 560  | 11 106 | 1 491  | 12,45     | 7,45      |
| Left ventricular maximal rate of pressure decline (dP/dt <sub>min</sub> , mmHg/s) | -17 616 | -7 133 | -1 869 | 9,43      | 3,82      |

➔ Most used models in physiology and pathophysiology, genetics, pharmacology and long-term survival studies

# Relevance of murine models

- Compared to larger animal models: Ease of handling, housing, shorter gestation time, lower maintenance cost
- Most widely used models in physiology and pathophysiology,
- Wide variety of study and analysis techniques (echocardiography, magnetic resonance imaging, electrocardiography, pressure-volume loop, blood pressure, and many others)
- Existence of numerous genetic models
- Numerous models of cardiac dysfunction and heart failure available
- Faster and less expensive proof-of-concept studies

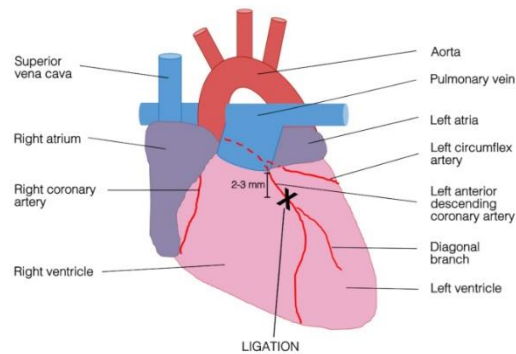
- Duration of the action potential is very different with absence of the plateau
- Mostly express  $\alpha$ MHC that have a faster kinetics but a higher energy cost than  $\beta$ MHC
- Predominant expression of the more rigid isoform titin (N2B) while N2A:N2B is more balanced in humans
- Degree of phosphorylation of troponin I (cTnI) is higher
- Calcium recapture is ensured at 90% by SERCA (only 76% in humans where NCX plays an important role)
- Very different functional parameters (see previous table)

# Models of ischemic heart failure\*

\*Heart failure= insuffisance cardiaque

## Surgical

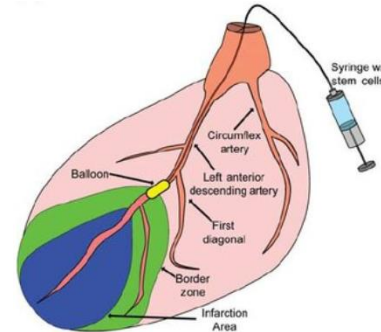
### Left Anterior Descending (LAD) Coronary Artery Ligation



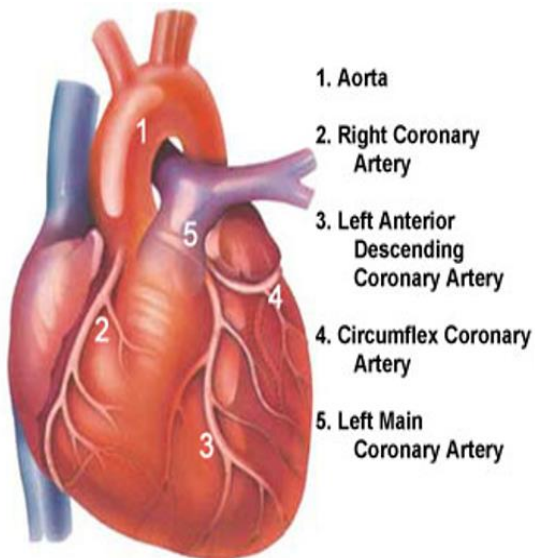
Jérôme Lugin et al., JOVE, 2019

## Endovascular

### Coronary occlusion with a balloon



Valina C. et al., Eur. Heart Journal, 2007

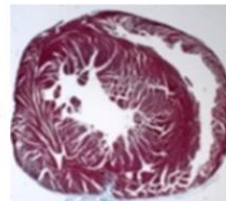


### Mouse

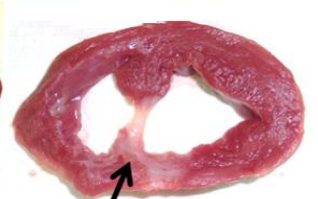
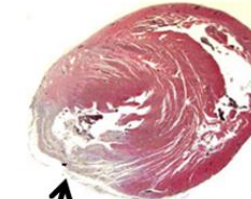
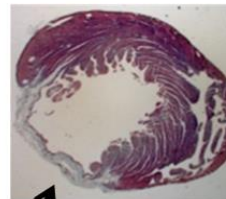
### Rat

### Pig

### Sham



### MI



Free wall

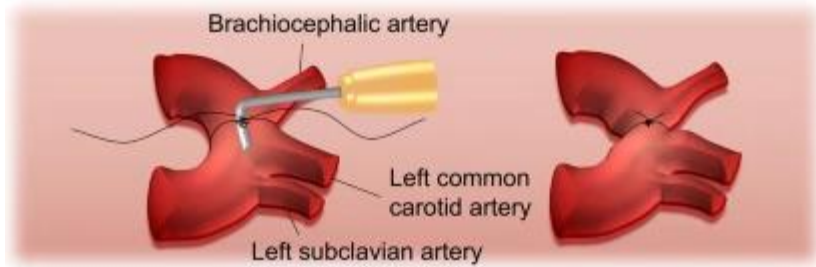
Free wall

IVS

# Non-ischemic models

## Surgical

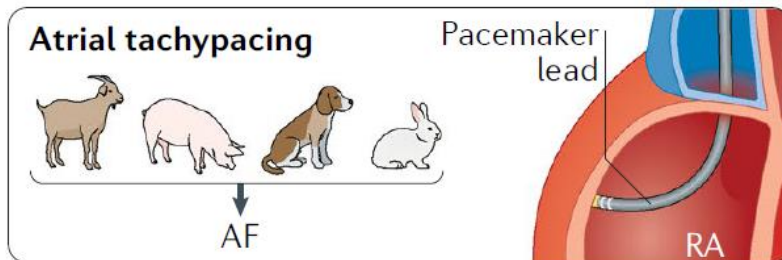
Aortic banding\*  
(model of hypertrophic cardiomyopathy)



Aortic banding\* = constriction aortique

Nakao Y. et al., Scientific Reports, 2022

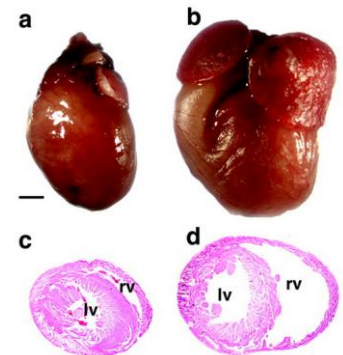
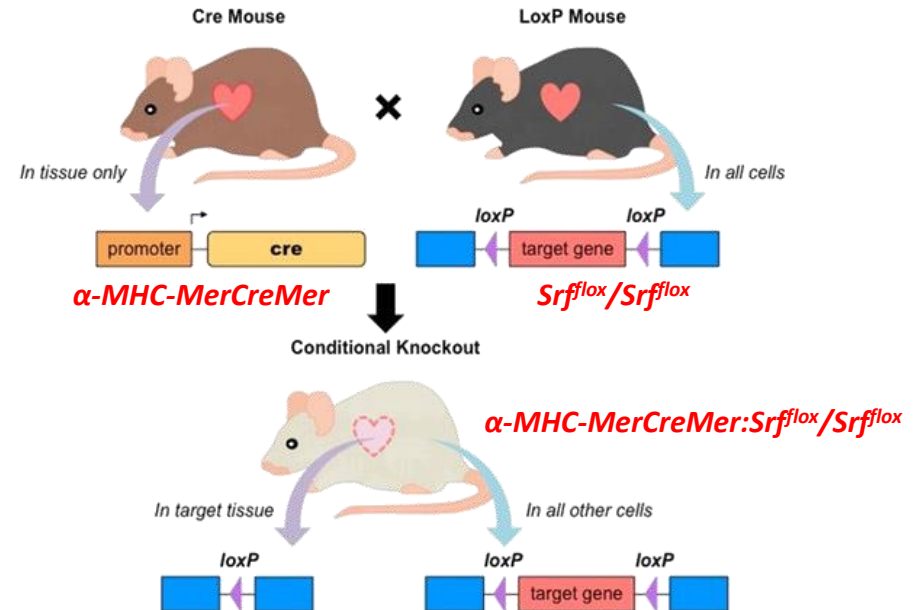
Atrial pacing  
(model of atrial fibrillation)



Clauss S. et al., Nat Rev Cardiol., 2019

## Transgenic

*Exemple of dilated cardiomyopathy*



Parlakian A. et al., Circulation., 2005

# Pour aller plus loin...

## Surgical Models

**Pro:**

- Very precise
- More variation between procedures
- Implantation of extra monitoring equipment

**Con:**

- High labor intensity
- Slow
- More post-OP complications
- More strain on animals

## Endovascular Models

**Pro:**

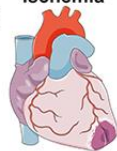
- Lower labor intensity
- Faster (allows for higher volume)
- Lower peri- / post-OP complications
- Less strain on animals

**Con:**

- Limited devices / methodologies
- Imaging required
- No implantation of extra equipment

### Ischemia

**Acute:** - Artery ligation ( $\pm$  reperfusion)



**Acute:** - Balloon occlusion ( $\pm$  reperfusion)  
- Embolization

**Chronic:** - Ameroid constrictor

**Chronic:** - Repeat microembolization  
- Bottleneck/Hourglass stent  
- Copper-plated stent

### Pressure Overload

- Aortic banding / cuffing  
- Banding of A. renalis  
- s.c. DOCA pellets (+diet)



- Stenting of young aorta desc.  
- Stenting of A. renalis

### Volume Overload

- Surgical mitral regurgitation  
- Arteriovenous fistula (AVF)



- Cutting mitral chordae

### Cytotoxic

- Doxorubicin (i.v.) + AVF



- Doxorubicin (intracoronary)

### Pacing

- Epicardial pacing leads  
- Single vs. Multiple leads



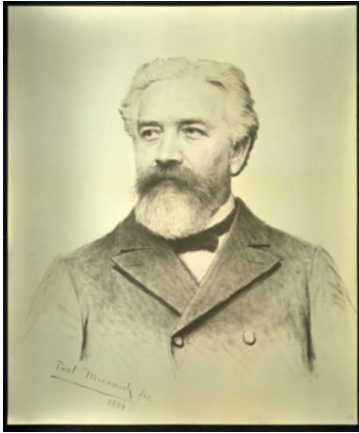
- Atrial vs. Ventricular pacing leads  
- Single vs. Multiple leads

| Type              | Method                                                     | Advantages                                                                                 | Disadvantages                                                                             | Literature                                                  |
|-------------------|------------------------------------------------------------|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| Acute ischemia    | Embolization                                               | Low-complexity, cheap                                                                      | Less reproducible, less precise, no reperfusion possible                                  | Canine (10)<br>Porcine (11–13)<br>Ovine (14, 15)            |
|                   | LAD ligation                                               | Reproducible, possibility of reperfusion, extra-coronary device implantation possible      | Open-chest, lethal arrhythmias, surgical complications, labor intensive,                  | Canine (16)<br>Porcine (17, 18)<br>Ovine (19, 20)           |
|                   | Balloon occlusion                                          | Reproducible, possibility of reperfusion, closed-chest, fewer complications                | Less precise than surgical ligation                                                       | Canine (21)<br>Porcine (22–27)<br>Ovine (28)                |
| Chronic ischemia  | Repeated microembolization                                 | Good model of microvascular disease                                                        | Repeat interventions, labor intensive                                                     | Canine (29, 30)<br>Ovine (31, 32)<br>Bovine (33)            |
|                   | Ameroid constrictor, bottleneck stent, copper-plated stent | Good model of single vessel disease                                                        | Variable degree of stenosis and LVEF reduction                                            | Canine (34)<br>Porcine (35–42)<br>Ovine (43)                |
| Pressure overload | Surgical aortic banding                                    | Reproducible, minimal surgical procedure                                                   | Abrupt increase in pressure                                                               | Canine (44)<br>Porcine (45, 46)<br>Ovine (47)               |
|                   | Stenting of descending aorta                               | Good model of aortic stenosis, gradual pressure overload, very similar to human conditions | Long time is necessary to develop left ventricular hypertrophy and dilation               | Porcine (48)                                                |
|                   | Renal artery stenosis                                      | Good model of severe systemic hypertension, low-complexity, single intervention            | Hormonal dysregulation, difficult to control                                              | Canine (49, 50)<br>Porcine (51)<br>Ovine (52)               |
|                   | DOCA pellet s.c. implantation with high salt diet          | Simple non-invasive method                                                                 | Mild systemic hypertension, uncertain effect on left ventricular function                 | Porcine (53)                                                |
| Volume overload   | Mitral regurgitation                                       | Good model of volume overload, reproducible                                                | Technically difficult                                                                     | Canine (54–57)<br>Porcine (58)<br>Ovine (59)                |
|                   | Arteriovenous fistula                                      |                                                                                            | Low clinical relevance, not well-established in large animals                             | Goat (60)<br>Ovine (61)                                     |
| Cytotoxic         | Anthracyclines                                             | Good model of doxorubicin-induced heart failure, simple non-invasive method                | Narrow clinical relevance, toxic substances, often only in combination with other methods | Canine (62, 63)<br>Porcine (64)<br>Ovine (65),<br>Goat (66) |
| Pacing            | Rapid pacing                                               | Good model of tachycardic remodeling                                                       | Reversible HF, qualitatively different remodeling                                         | Canine (66, 67)<br>Porcine (68–72)<br>Ovine (73)            |

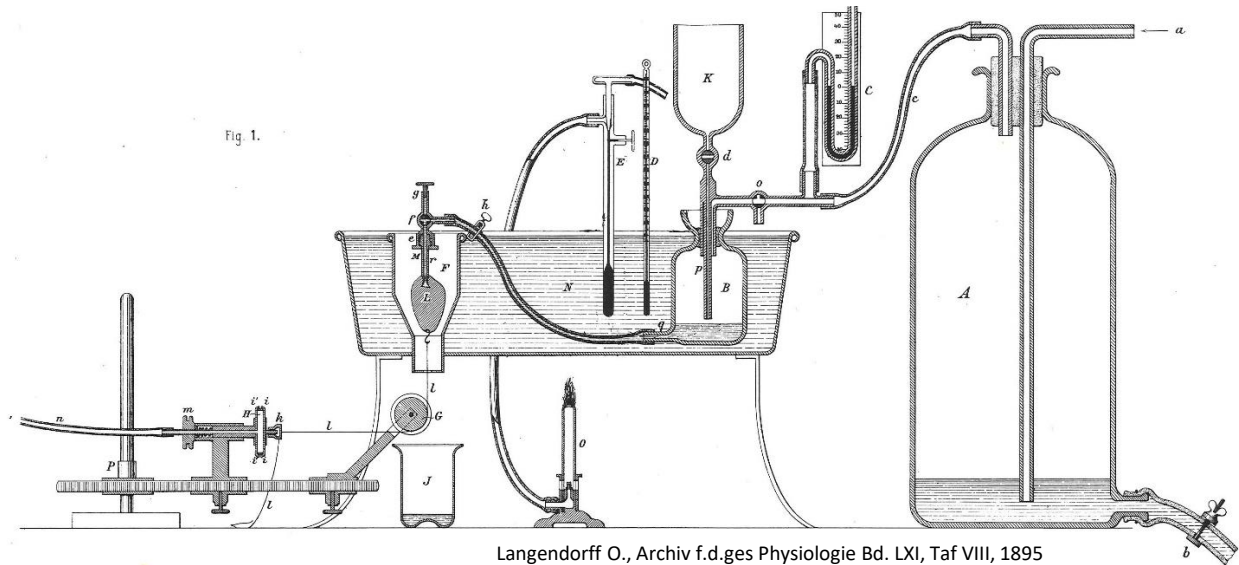
Summarizes the different kinds of HFREF models, their main uses and their advantages and disadvantages. The selected papers are meant to serve as illustrative examples of the various models, they do not represent a comprehensive list of all studies employing the respective model. LAD, Left anterior descending; LVEF, Left ventricular ejection fraction; DOCA, deoxycorticosterone acetate; sc, subcutaneous; HF, Heart Failure.

### **III. *Ex vivo* models**

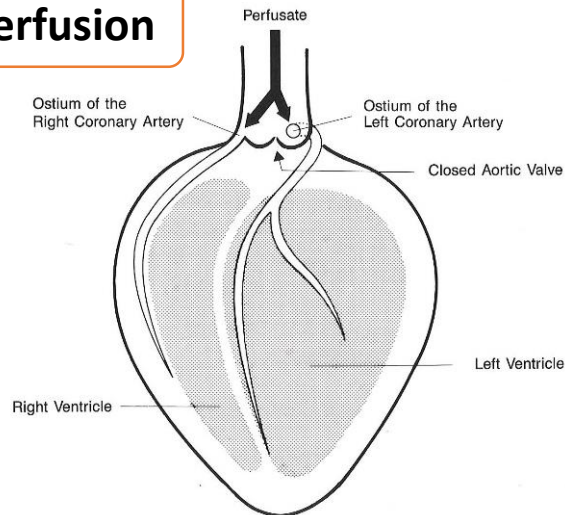
# The isolated perfused heart according to Langendorff



Oscar Langendorff, 1853-1908

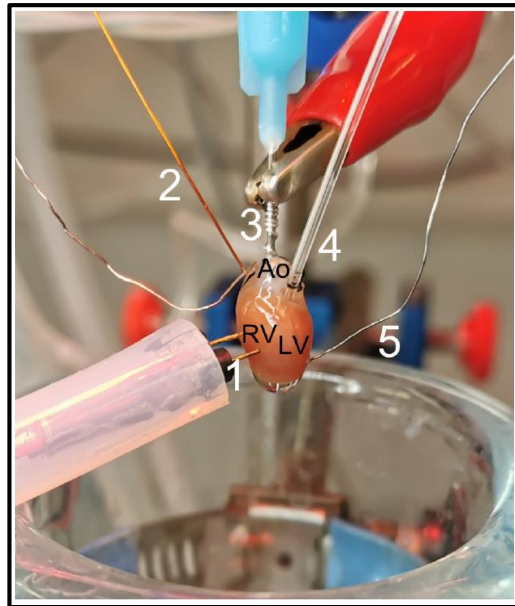
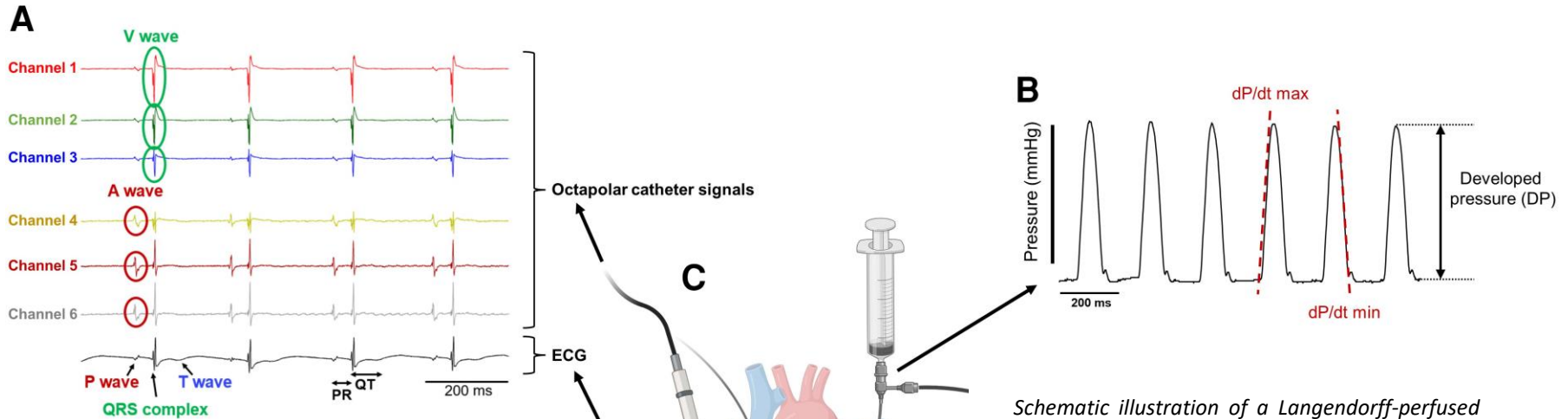


## The principle: retrograde perfusion



Perfusion paths at the LANGENDORFF-heart, very simplified portrayal (ventral view of the heart). The aortic valve is closed; right and left ventricle remain empty. The perfusate leaves the coronary vascular system at the coronary sinus (not shown) and drips off at the heart's apex.

# A model still in use: an example in mice



1= pacing electrodes, 2 = intracardiac octapolar catheter, 3 = aortic cannula, 4 = intraventricular balloon system coupled to pressure measurement, 5 = silver wires, Ao, aorta, RV, right ventricle, LV, left ventricle.

## Electrical parameters

- heart rate
- PR interval
- QT interval
- intra-atrial and intraventricular conduction times
- ventricular effective refractory period

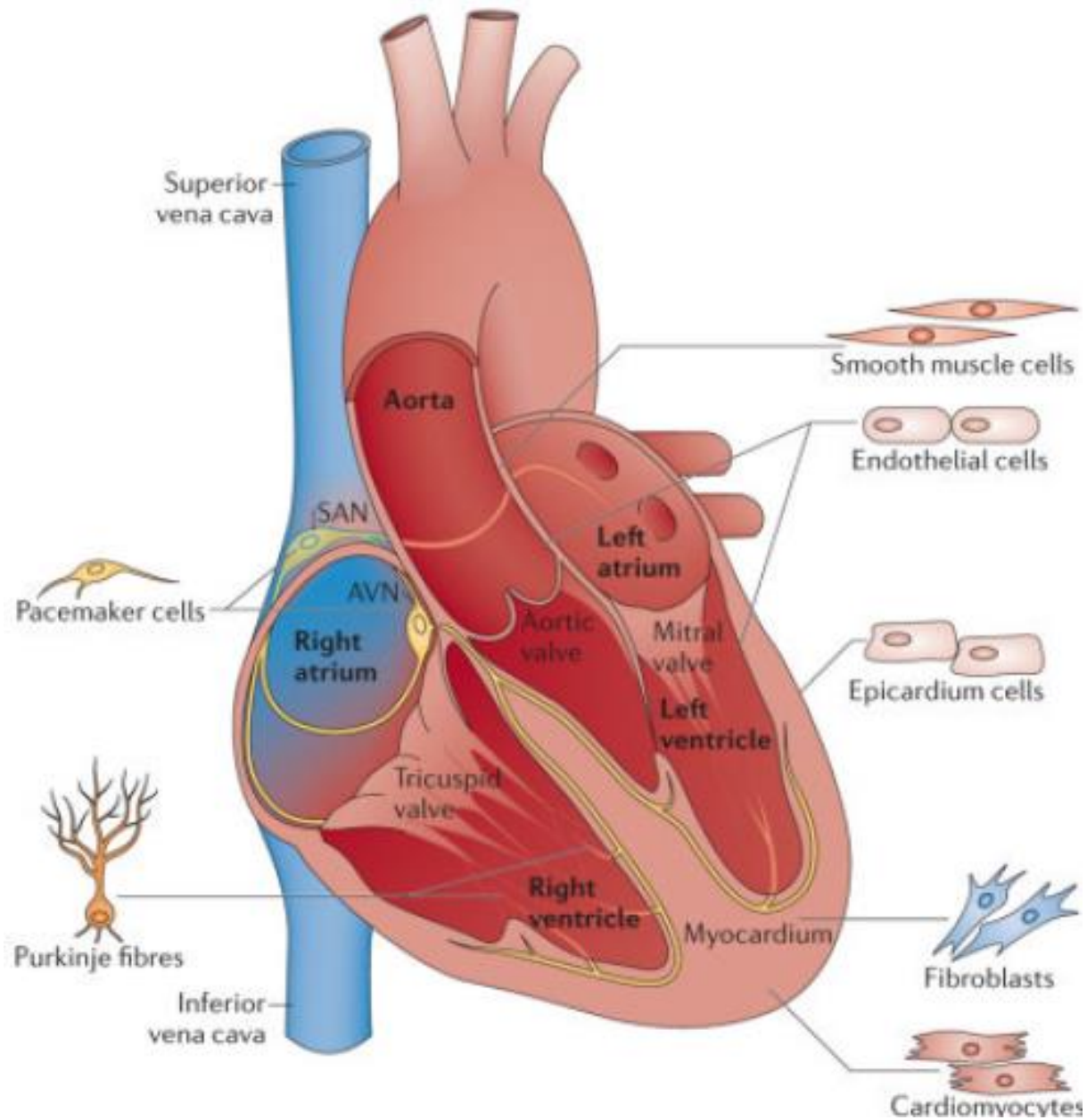
## Mechanical parameters

- developed pressure (DP)
- left ventricular (LV) work
- maximal velocity of contraction (dP/dt max)
- maximal velocity of relaxation (dP/dt min)

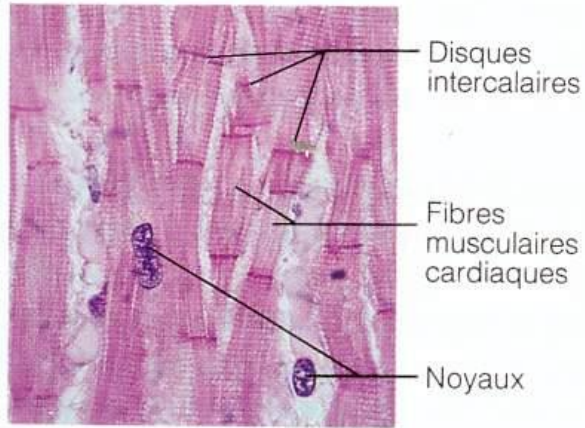
*Schematic illustration of a Langendorff-perfused heart with the intracardiac octapolar catheter and corresponding electrodes 1–8 (Ch = channel) inserted in the right atrium and ventricle (1), silver wires to record the electrocardiogram (2), pacing electrodes to assess the ventricular effective refractory period (3), and the balloon (4) inserted in the left ventricle coupled to pressure recording (5) to measure left ventricular pressure.*

## IV. Cell models (*in vitro*)

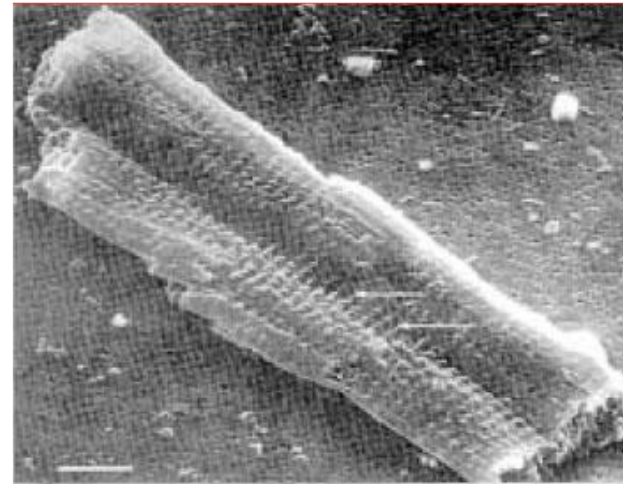
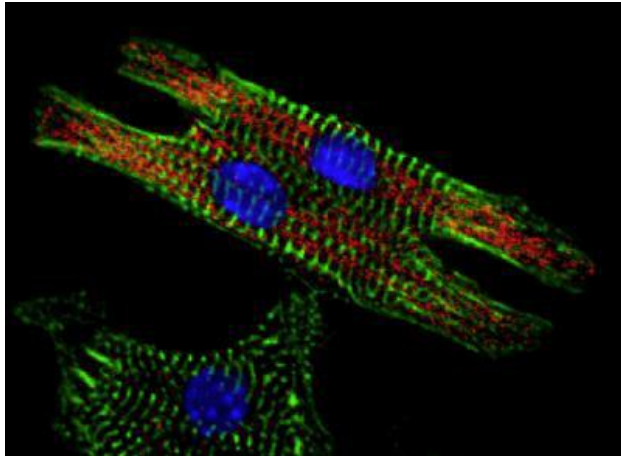
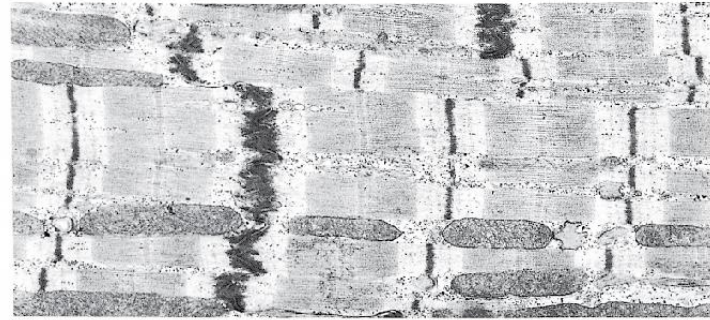
# The cardiac cells



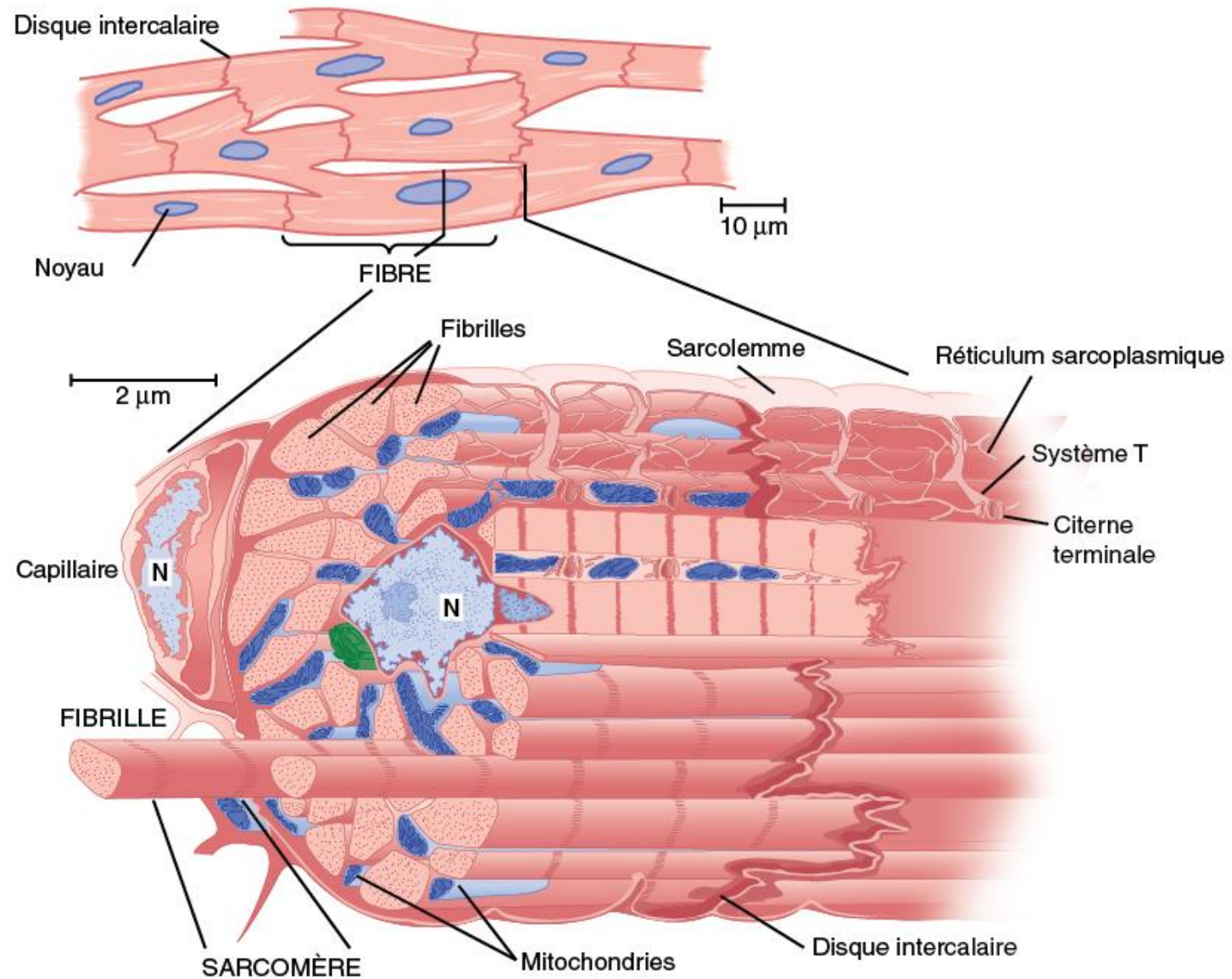
# Cardiomyocytes: functional units of the heart



A



# Structures of cardiomyocytes



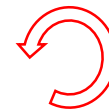
# Cells models for cardiac « physiology »

|             | Neonatal cardiomyocytes                                                                                                                                                                                                                    | Adult cardiomyocytes                                                                                                                                                                                                                                    | Cardiac cell lines                                                                                                                                                                                                                                        | ESC/iPSCs                                                                                                                                                                                     |
|-------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Pros        | <ul style="list-style-type: none"> <li>• Easy primary isolation</li> <li>• Able to culture for a long period</li> <li>• Well-established chemical drugs to induce pathologic conditions</li> <li>• High transfection efficiency</li> </ul> | <ul style="list-style-type: none"> <li>• High similarity to human cells in the morphology and behavior</li> <li>• Able to obtain cells from diseased heart</li> <li>• Able to assess contractility and <math>\text{Ca}^{2+}</math> transient</li> </ul> | <ul style="list-style-type: none"> <li>• Easy to culture</li> <li>• Able to culture for a long period</li> <li>• Able to passage and recover from a frozen stock</li> <li>• High transfection efficiency</li> <li>• Minimize animal sacrifices</li> </ul> | <ul style="list-style-type: none"> <li>• Derived from human source</li> <li>• Able to assess <math>\text{Ca}^{2+}</math> transient</li> <li>• Able to study human genetic disorder</li> </ul> |
| Cons        | <ul style="list-style-type: none"> <li>• Immaturity</li> <li>• Non-uniform contraction</li> </ul>                                                                                                                                          | <ul style="list-style-type: none"> <li>• Limited gene manipulation methods</li> <li>• Technically difficult to isolate cells</li> </ul>                                                                                                                 | <ul style="list-style-type: none"> <li>• Lack of cardiac phenotype</li> <li>• Unable to assess contractility</li> </ul>                                                                                                                                   | <ul style="list-style-type: none"> <li>• Technically difficult</li> <li>• High cost</li> <li>• Immaturity</li> <li>• Batch-to-batch variation</li> </ul>                                      |
| Application | <ul style="list-style-type: none"> <li>• Hypertrophy assessment</li> <li>• High-throughput drug screen</li> </ul>                                                                                                                          | <ul style="list-style-type: none"> <li>• Contractility measurement</li> <li>• <math>\text{Ca}^{2+}</math> imaging</li> <li>• Patch-clamp</li> <li>• Sublocalization study by immunostaining</li> </ul>                                                  | <ul style="list-style-type: none"> <li>• High-throughput drug screen</li> <li>• Toxicology</li> </ul>                                                                                                                                                     | <ul style="list-style-type: none"> <li>• High-throughput drug screen</li> <li>• Toxicology</li> <li>• <math>\text{Ca}^{2+}</math> imaging</li> <li>• Precision medicine</li> </ul>            |

In human?

# Pluripotent stem cells

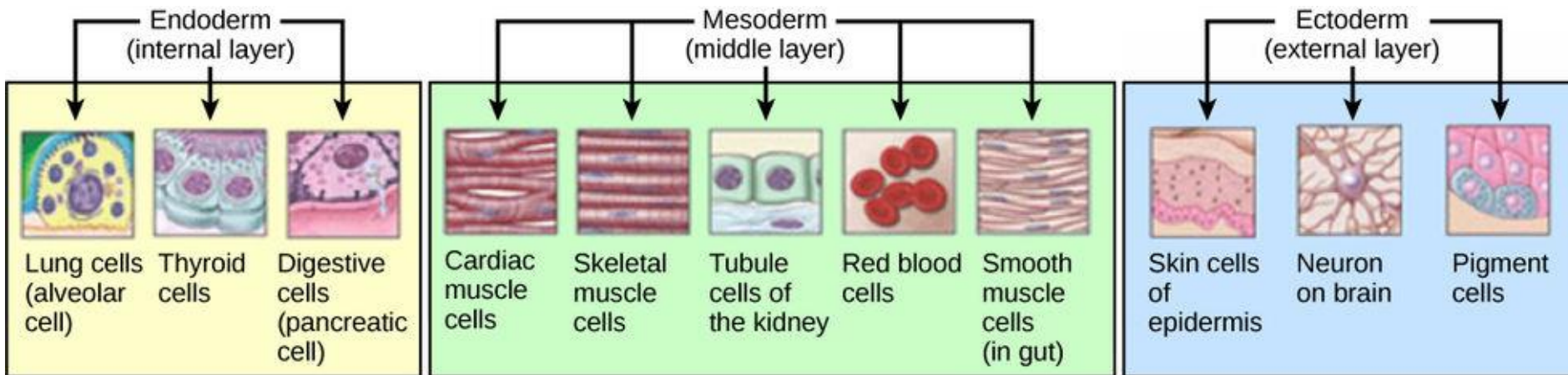
Pluripotent stem cells



Self-renewal



Differentiation



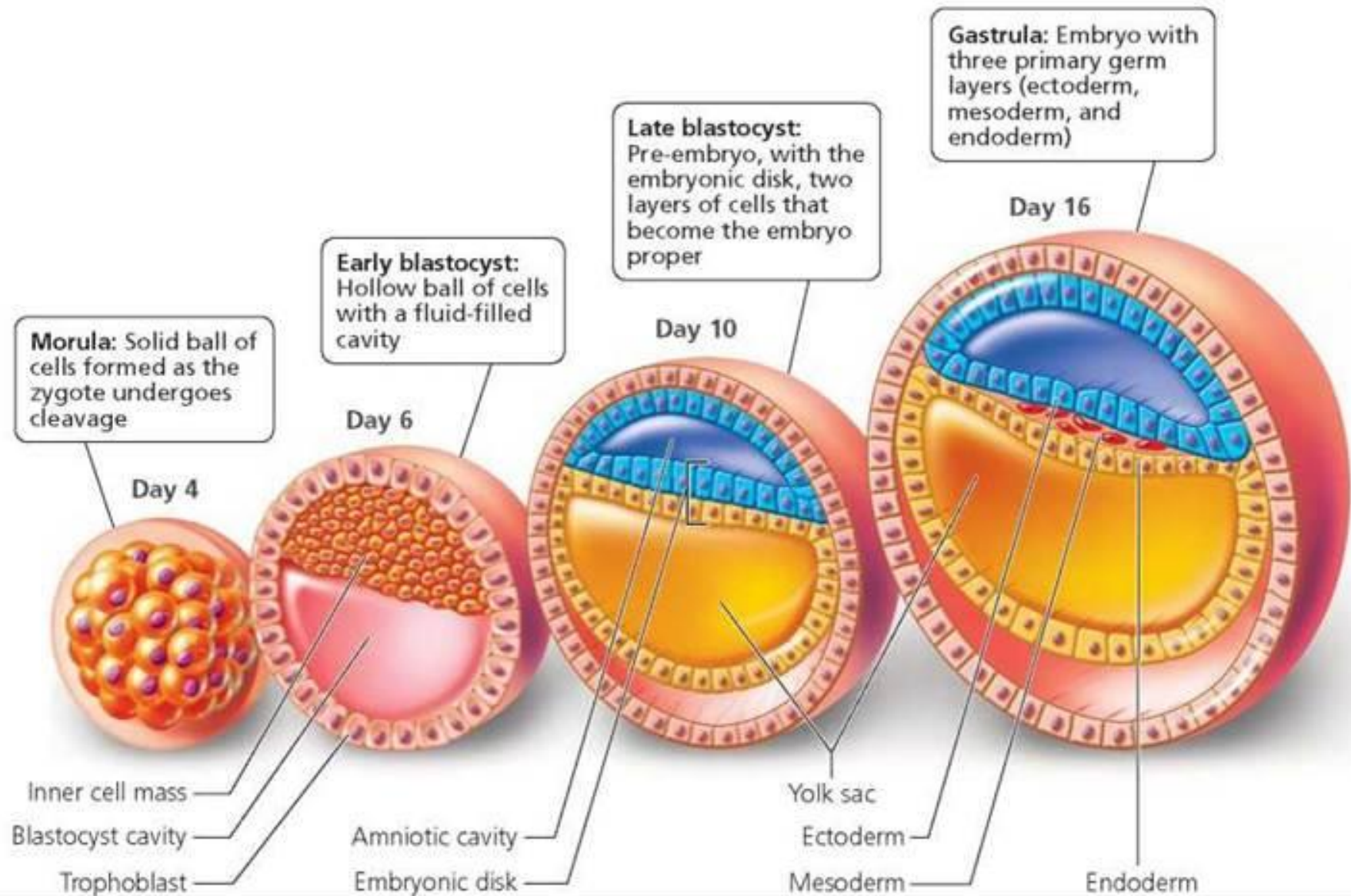
**Self-renewal** - ability to undergo numerous cycles of cell division while maintaining the undifferentiated or unspecialized state

**Clonality** – ability of a single cell to form many similar cells

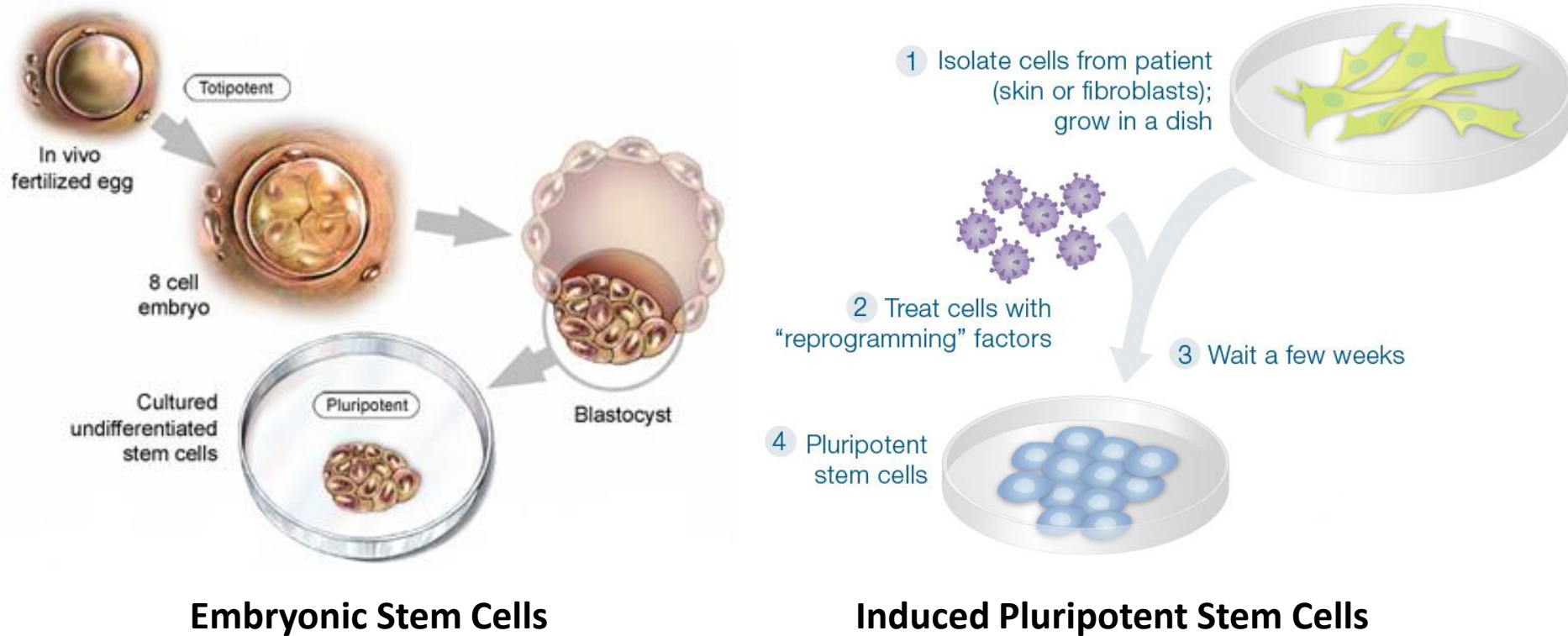
**Differentiation** - process by which a less specialized cell becomes a more specialized one.

**Potency**- the potential for differentiation to specialized cell types

# Origins of ESCs



# Comparison ES/iPS



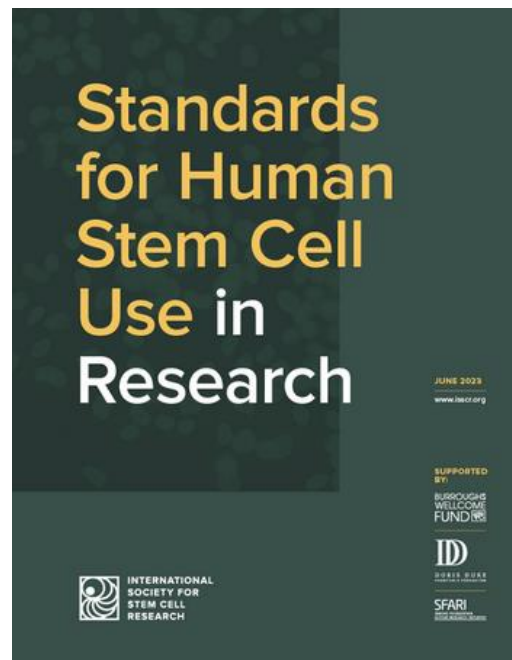
- ✓ Close characteristics between ESC and iPSC
- ✓ Pluripotent = can differentiate in all cell type of the organism (ectoderm, mesoderm, endoderm)
  - ✓ Self-renewal

*Self-renewal is the process of giving rise to indefinitely more cells of the same cell type.*

# Guidelines to derive and characterize stem cell lines

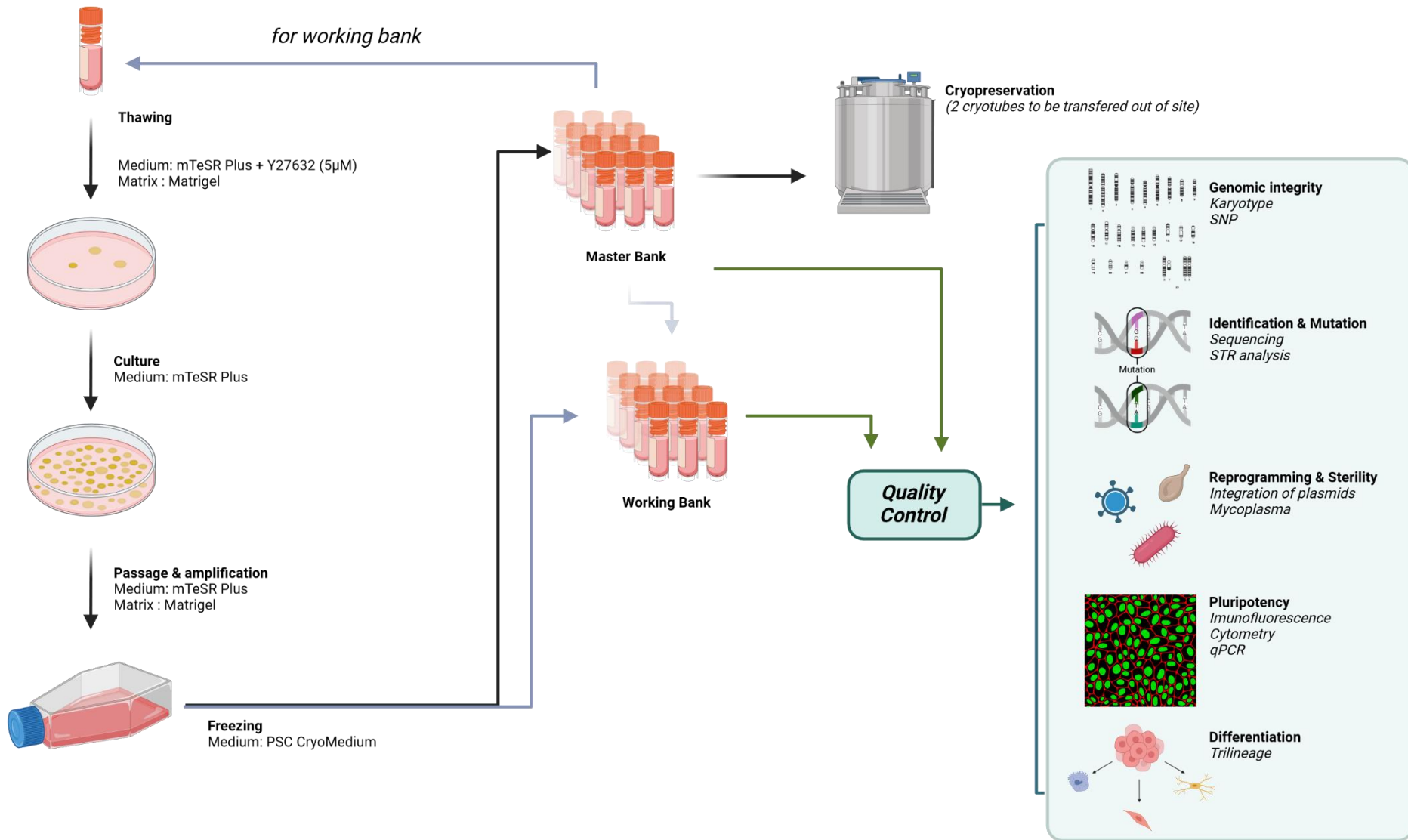


INTERNATIONAL  
SOCIETY FOR  
STEM CELL  
RESEARCH

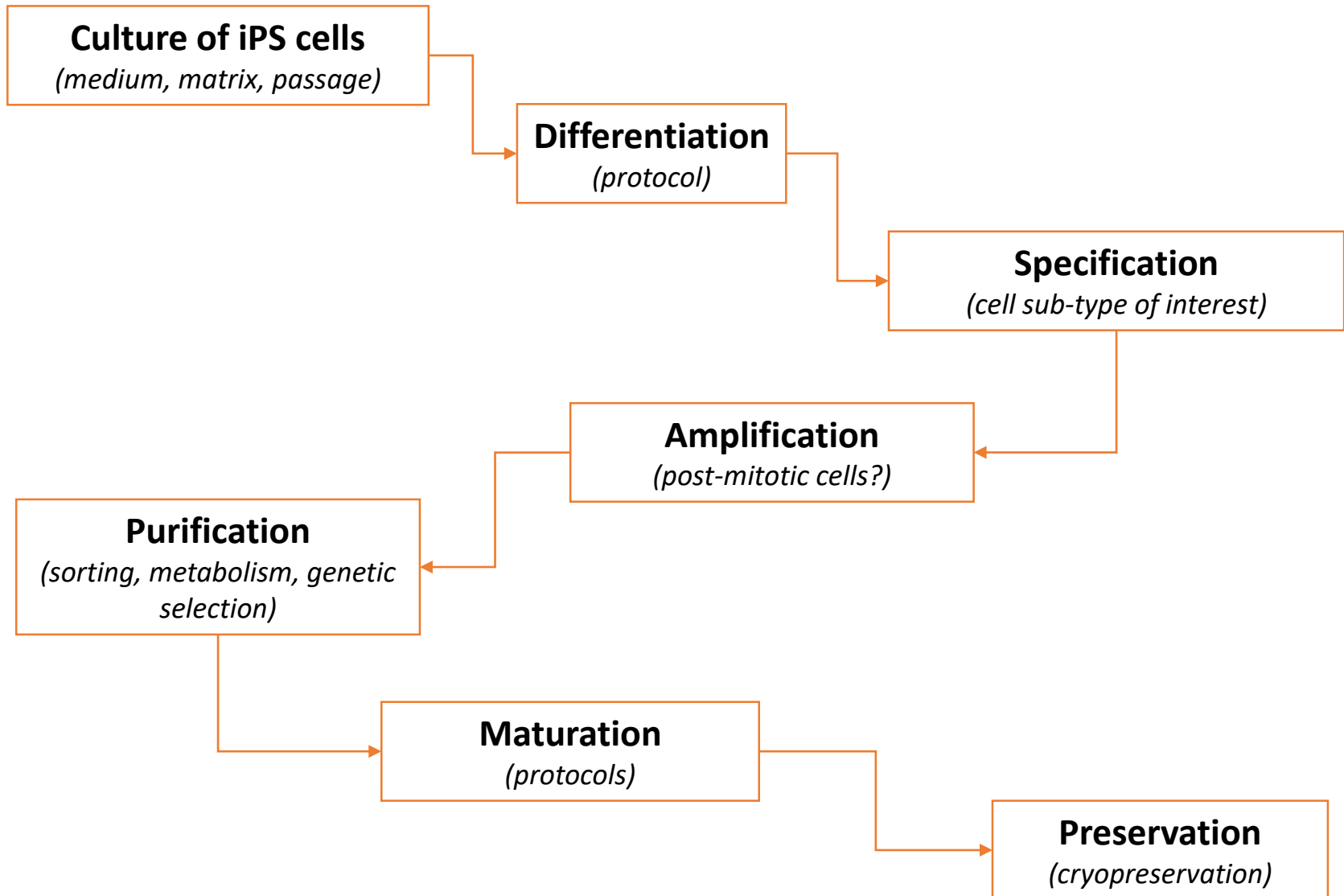


**Lab resource  
publications**  
*(hiPSC lines)*

# A workflow to bank and control hiPSCs



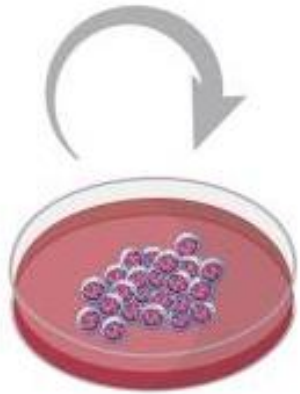
# Differentiation of pluripotent stem cells: general workflow



# Cardiac differentiation

## Culture of iPS cells

(medium, matrix, passage)



Matrigel or Synthetmax  
Media: E8 or B8

## Differentiation

(factors of differentiation, cellular modalities)

Checkpoints:

- Confluence
- Morphology

Procedures:

- Aggregation
- Replating

### Cytokines

(BMP4, actinin A...)

### Molecules

(CHIR99021, IWP2)



Monolayer



Embryoid Body

## Specification

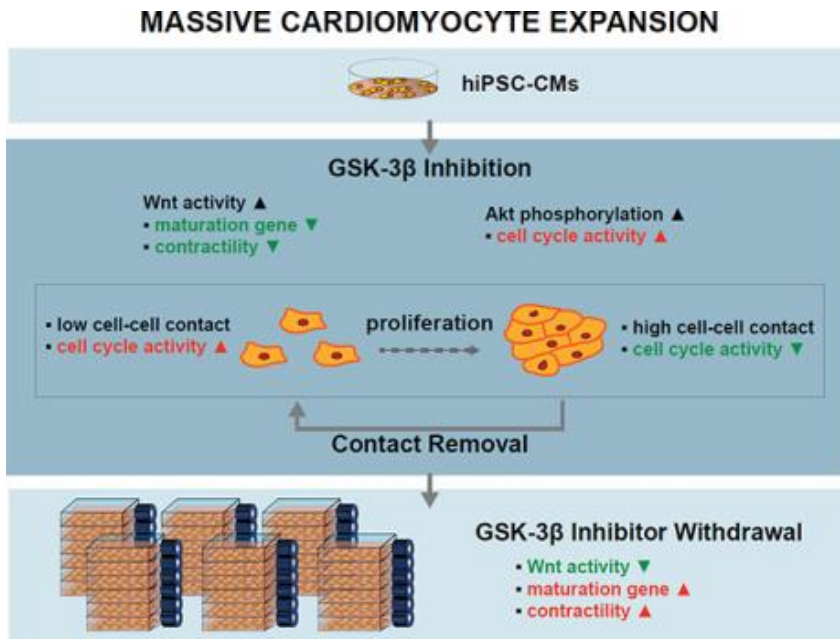
(cell sub-type of interest)

|                        | day 0                                 | day 1 | day 2                   | day 3    | day 4                        | day 5           | day 6                   | day 7 |                              |
|------------------------|---------------------------------------|-------|-------------------------|----------|------------------------------|-----------------|-------------------------|-------|------------------------------|
|                        | Mesoderm & cardiac mesoderm induction |       | Early cardiac induction |          | Cardiac progenitor formation |                 | Cardiomyocyte formation |       |                              |
| Sub-type specification | CHIR99021                             |       | Wnt-C59                 |          |                              |                 |                         |       | Burrige et al., 2014         |
|                        | Ventricular                           |       | CDM3                    |          |                              |                 |                         |       |                              |
|                        | CHIR99021                             |       | IWP-2                   | IWP-2 RA | RA                           | RA              |                         |       | Kleinsorge and Cyganek, 2020 |
|                        | Atrial                                |       | CDM3                    |          |                              |                 |                         |       |                              |
|                        | CHIR99021                             |       |                         | IWP-2    |                              | CHIR99021 WNT5B |                         |       | Ren et al., 2019             |
|                        | Nodal                                 |       | RPM1/B27                |          |                              |                 |                         |       |                              |

# Amplification and purification

## Amplification

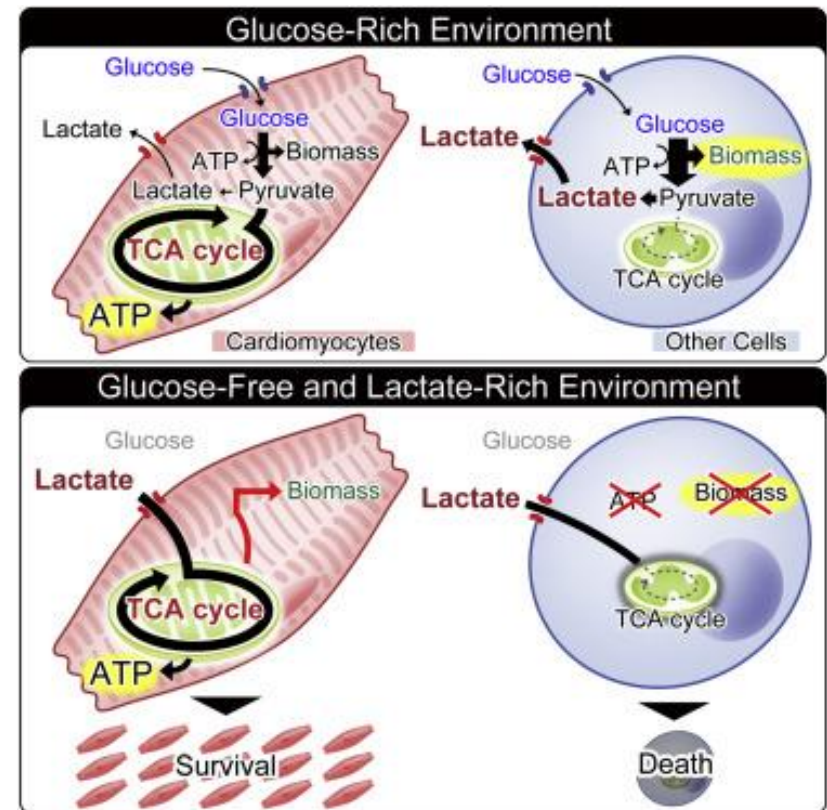
(density and inhibition of GSK3beta)



Jan W. Buikema et al., Cell Stem Cell, 2020

## Purification

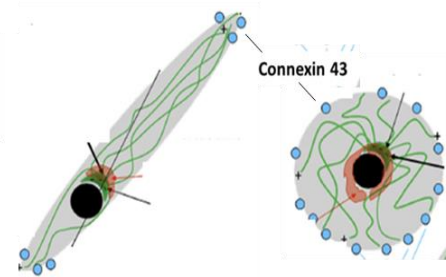
(metabolic)



Shugo Tohyama et al., Cell Stem Cell, 2013

# Immaturity of hiPSC-CMs

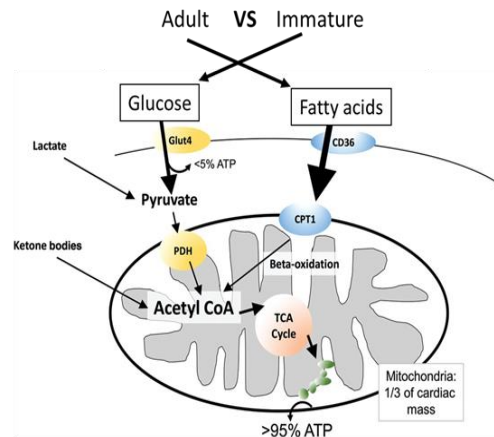
## Morphology



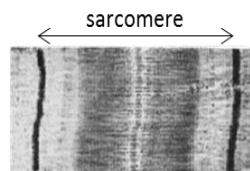
Adult VS Immature

|                            |               |                    |
|----------------------------|---------------|--------------------|
| Shape                      | Rod/Elongated | Circular/Polygonal |
| T-tubules                  | Present       | Absent             |
| Multicellular organization | Polarized     | Disorganized       |
| Binucleation               | In 25-30%     | Sporadically       |

## Metabolism



## Contractile unit

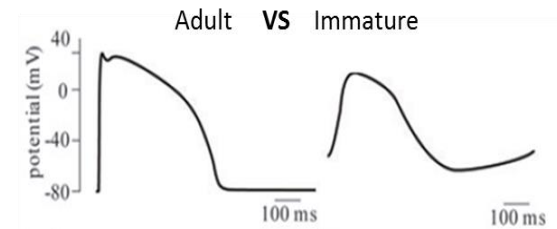


Adult VS Immature

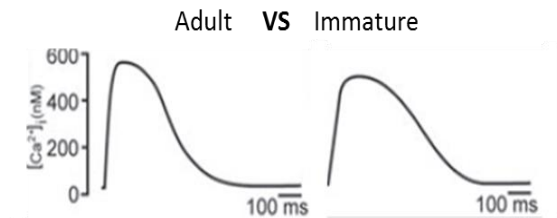
|           |           |              |
|-----------|-----------|--------------|
| Sarcomere | Organized | Disorganized |
| Size      | 150 pF    | 20-30 pF     |

## Electrophysiology

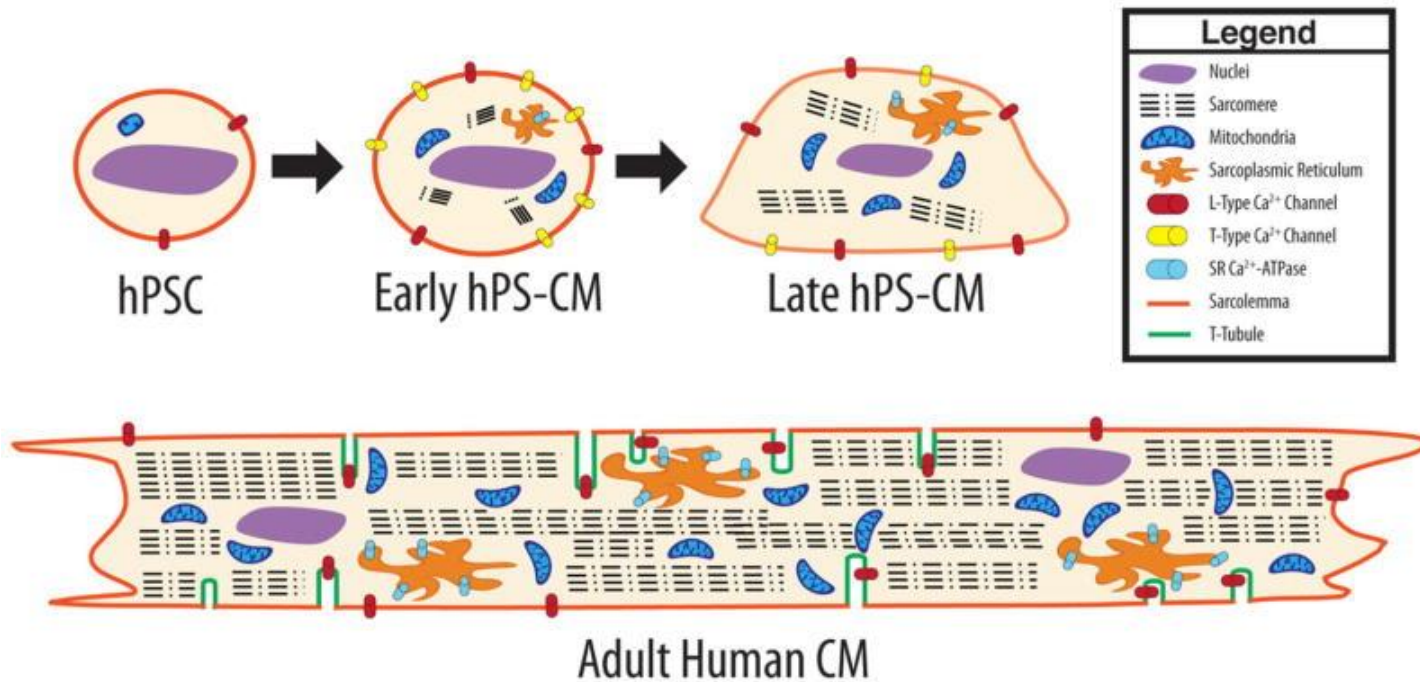
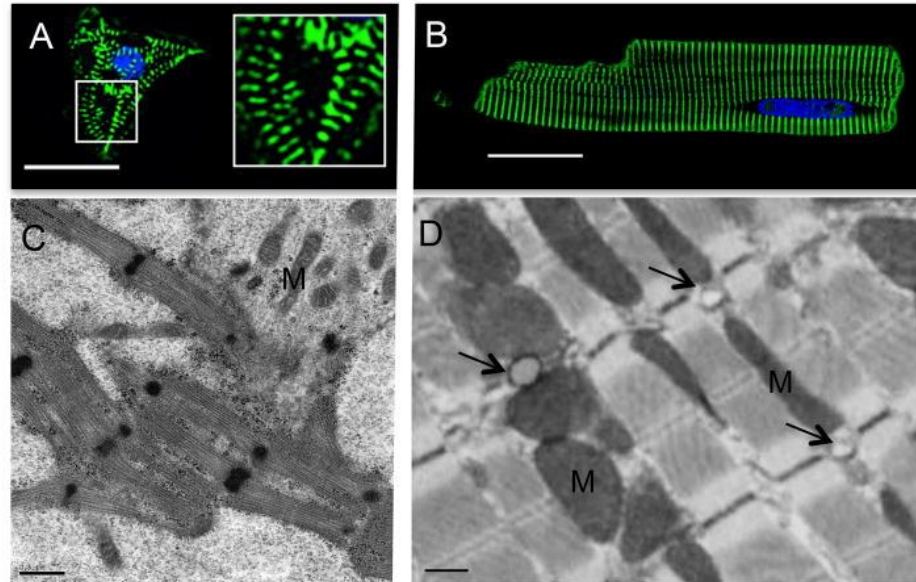
### Action potential



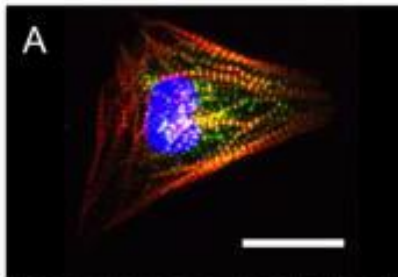
### Calcium handling



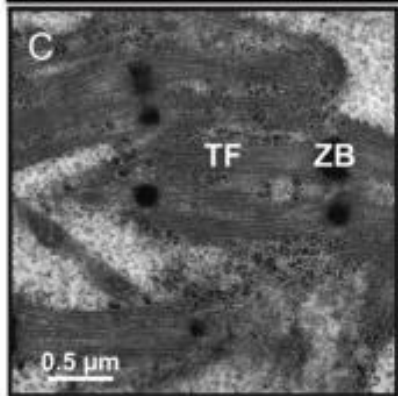
# Example : structural immaturity



# Maturation strategies

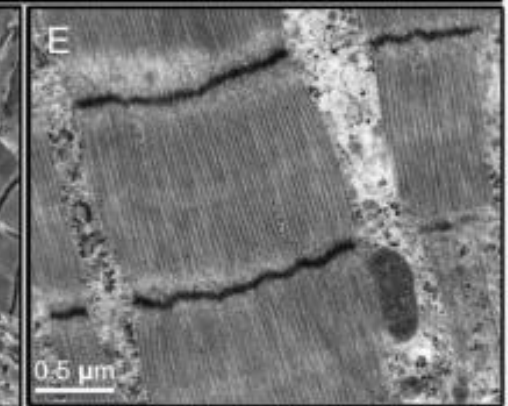
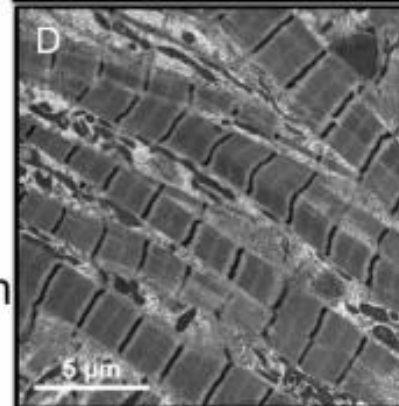
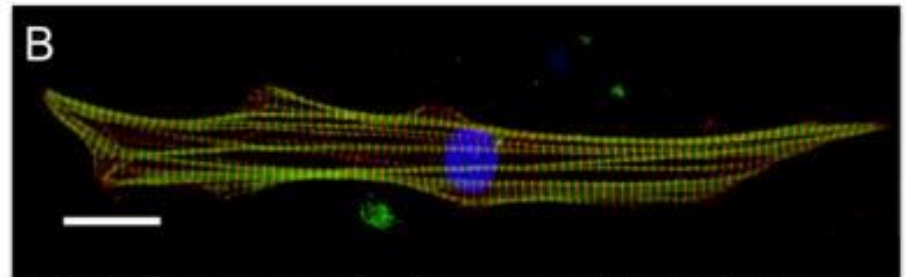


Long-term Culture  
Substrate Stiffness  
Cell Patterning



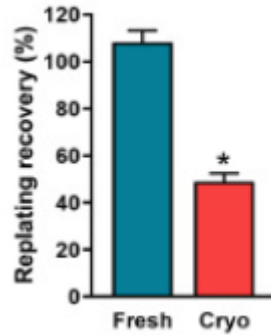
Biochemical Cues

Tissue Engineering  
Mechanical Loading  
Electrical Stimulation

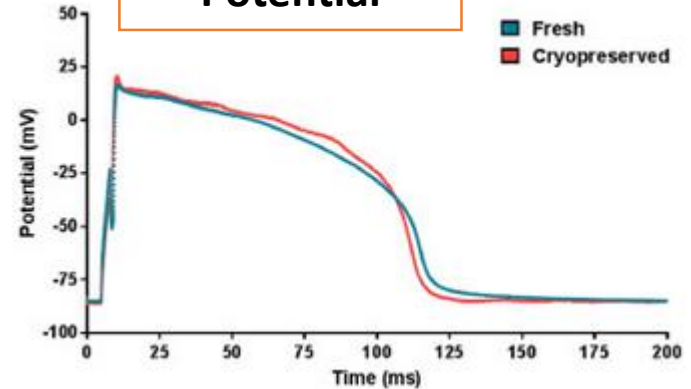


# Cryopreservation of hiPSC-CMs

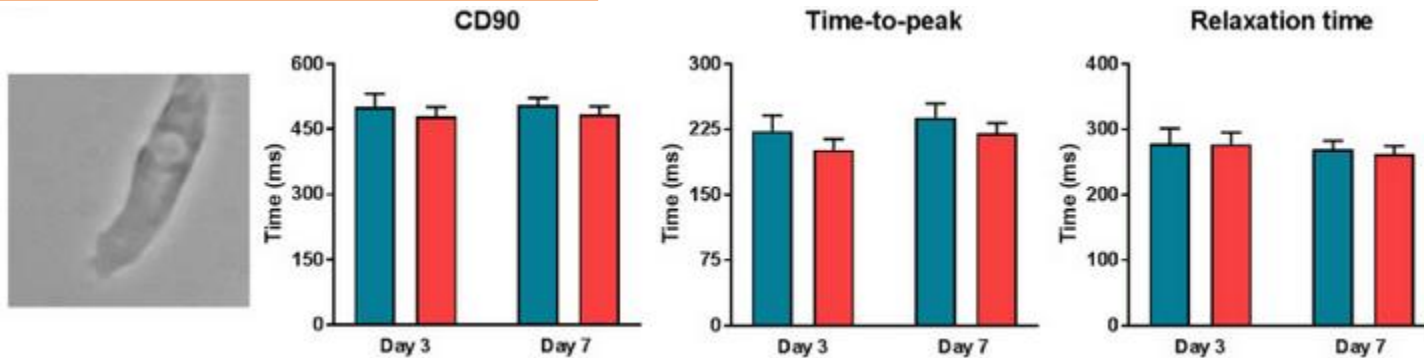
## Survival



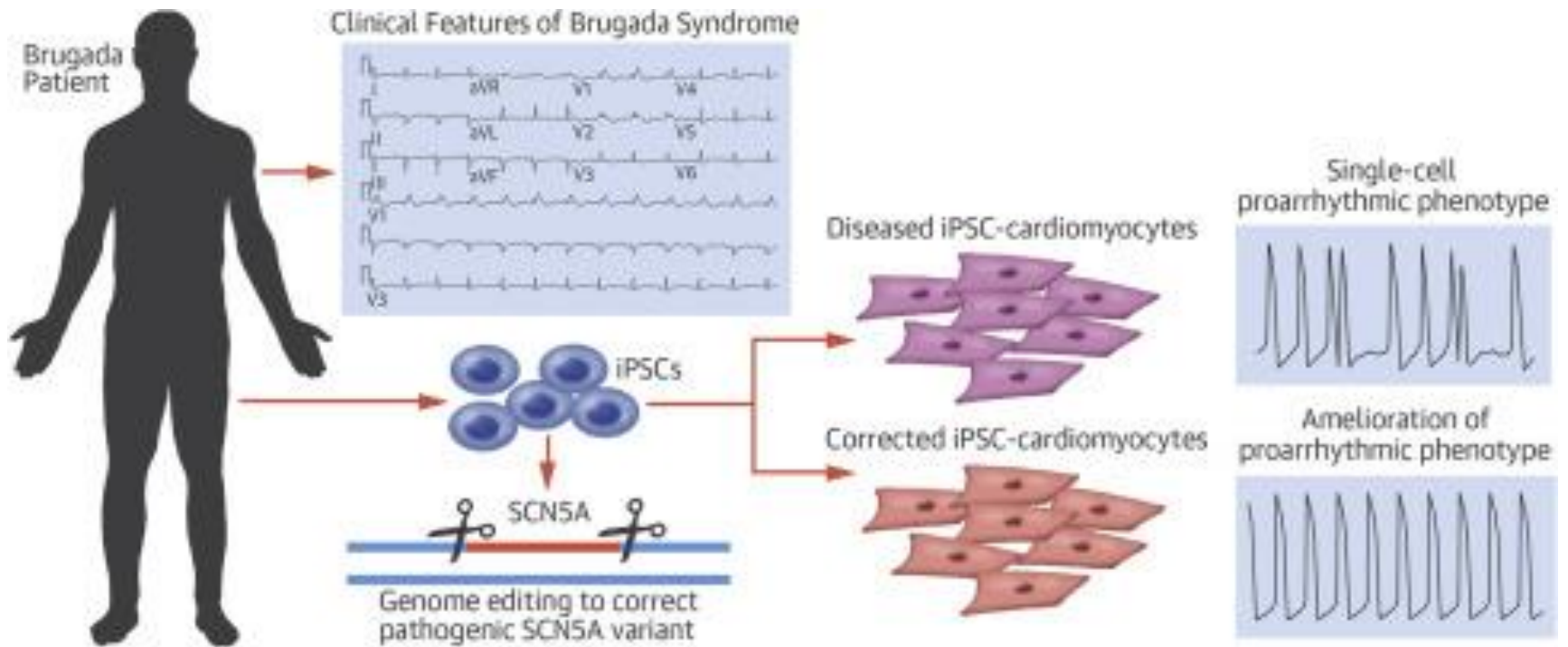
## Action Potential



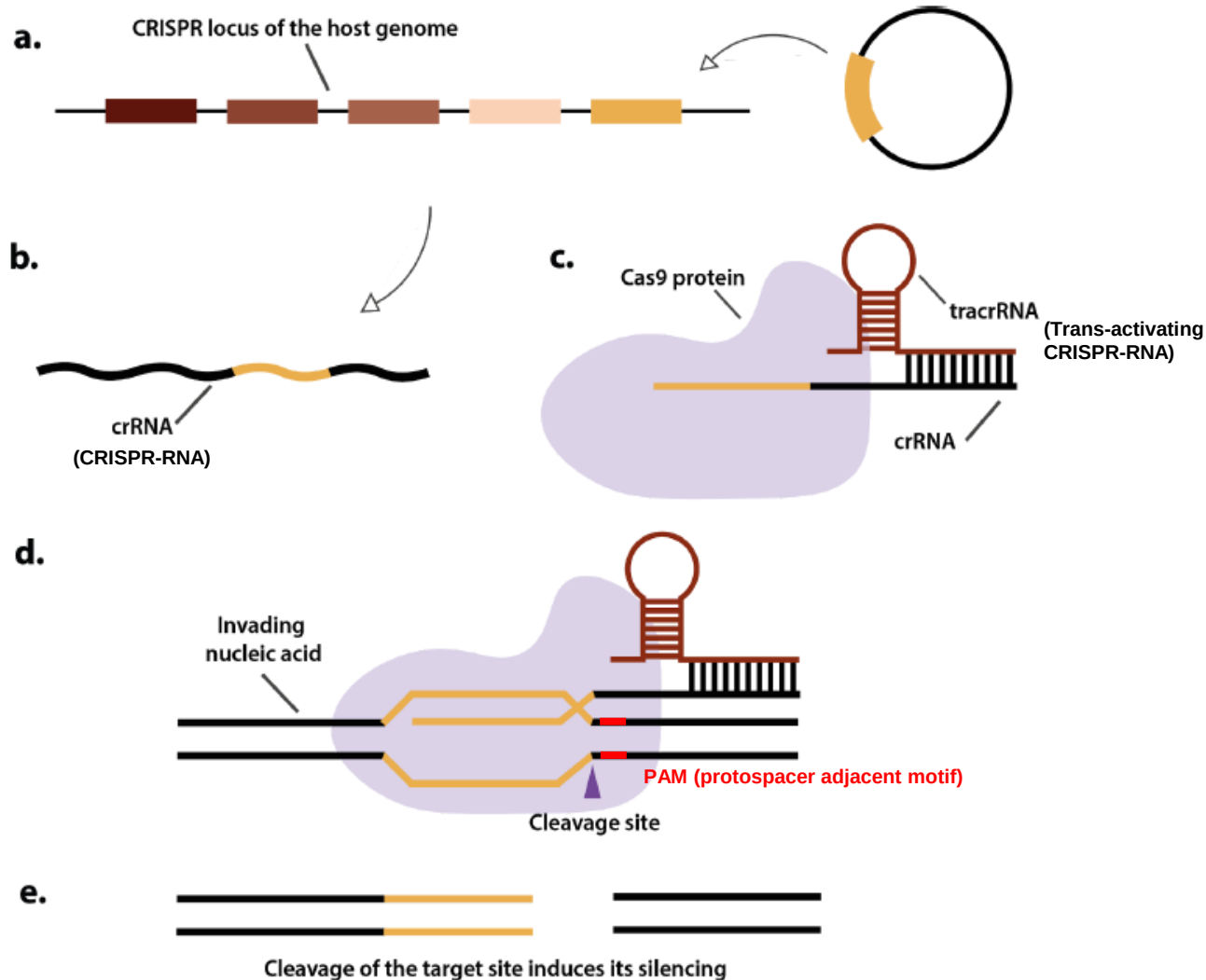
## Contractile Parameters



# Generate a control cell line



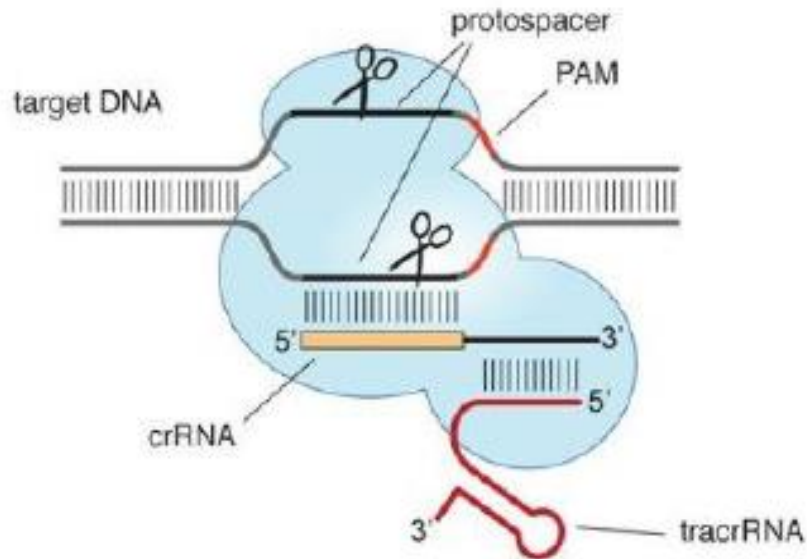
# CRISPR/Cas9 – A defense system in bacteria



CRISPR: Clustered  
Regularly Interspaced  
Palindromic Repeat

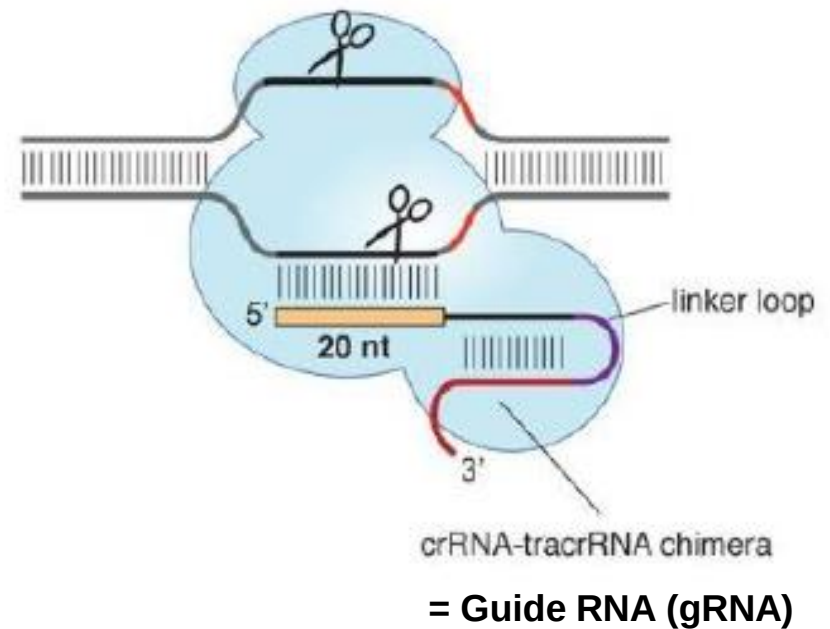
# A tool for genome editing

Cas9 programmed by crRNA:tracrRNA duplex

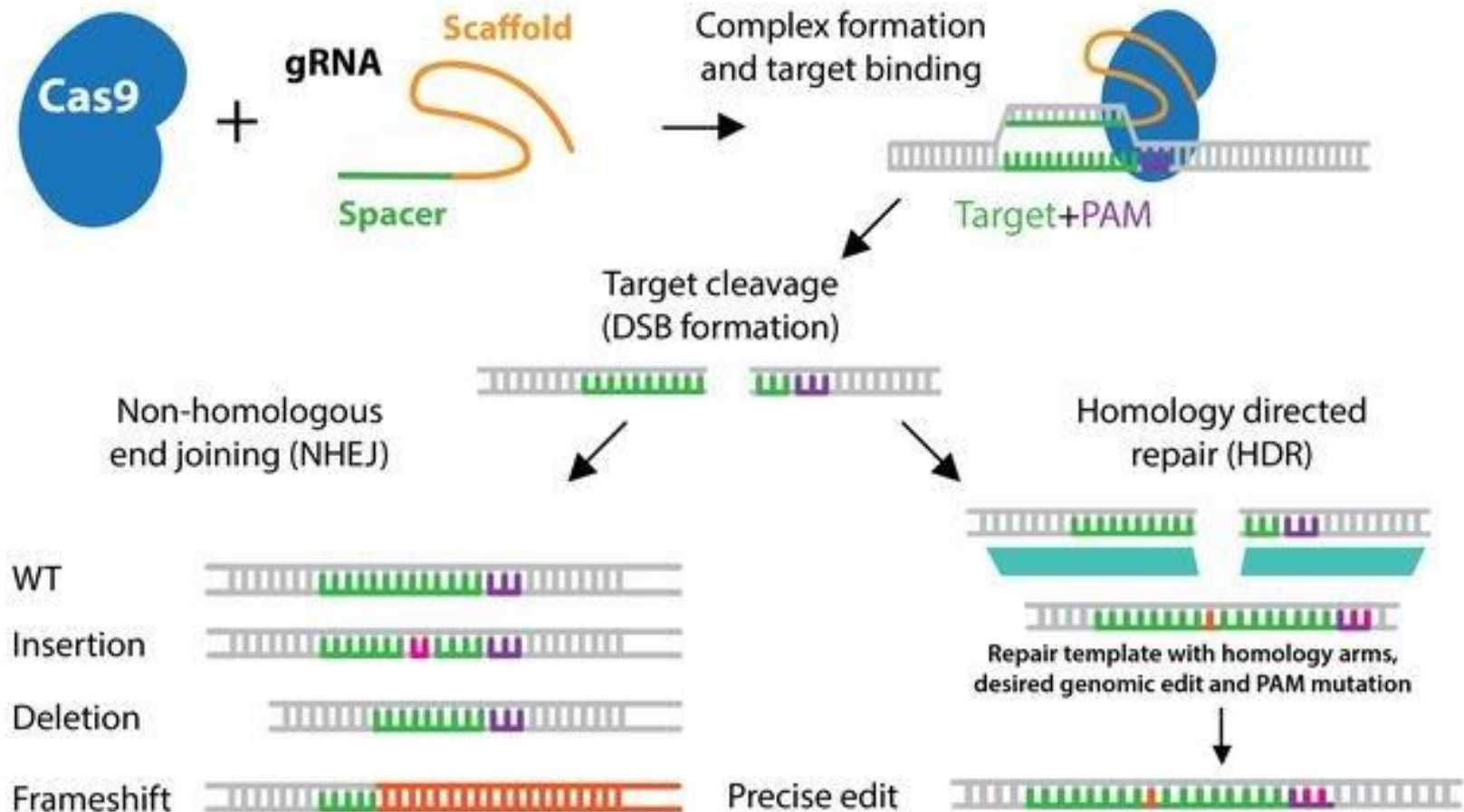


From Jinek et al. 2012

Cas9 programmed by single chimeric RNA

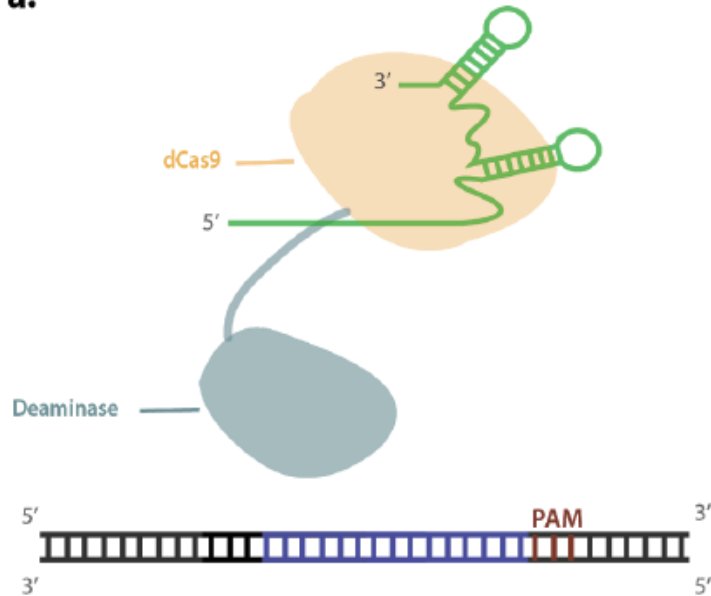


# DNA repair following Cas9 cleavage

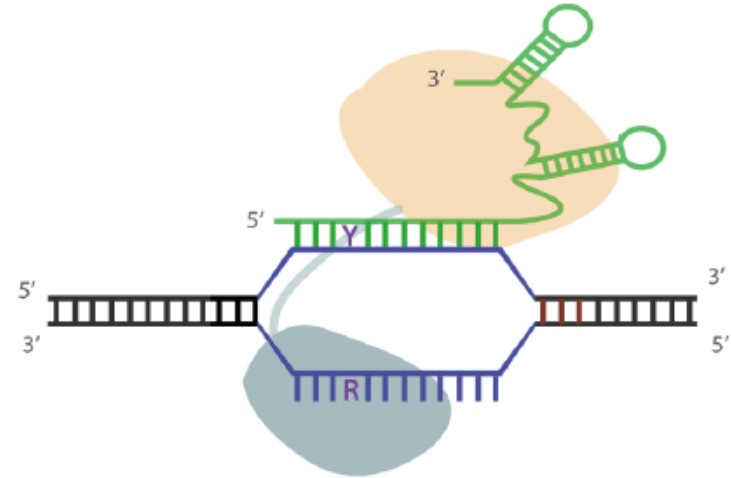


# Base editing

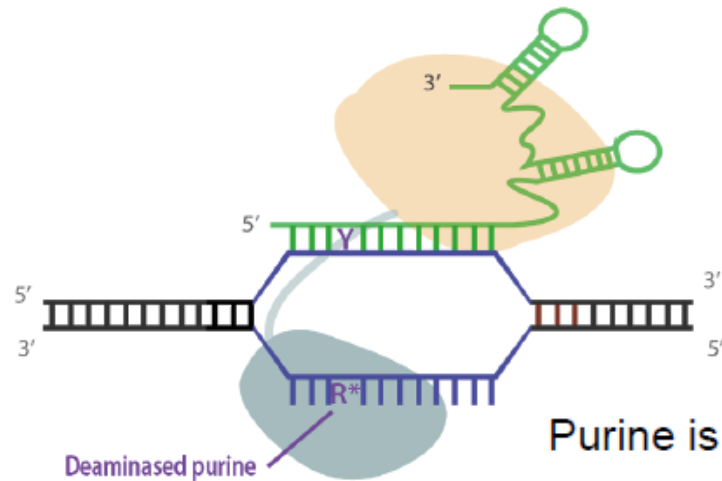
**a.**



**b.**



**c.**



**d.**



Purine is deaminated



C → U → read as a T  
A → I → read as a G

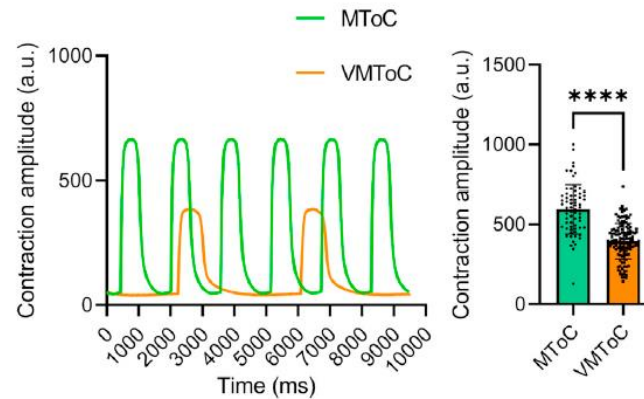
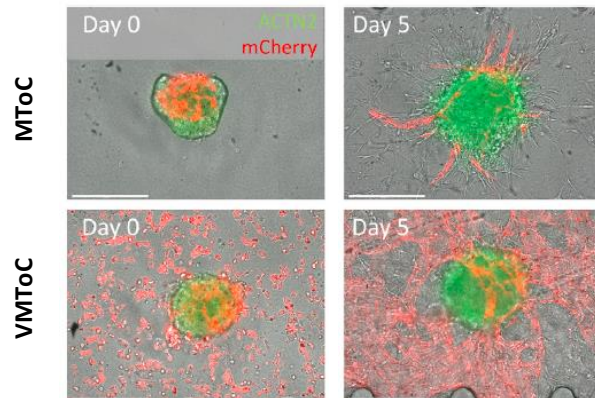
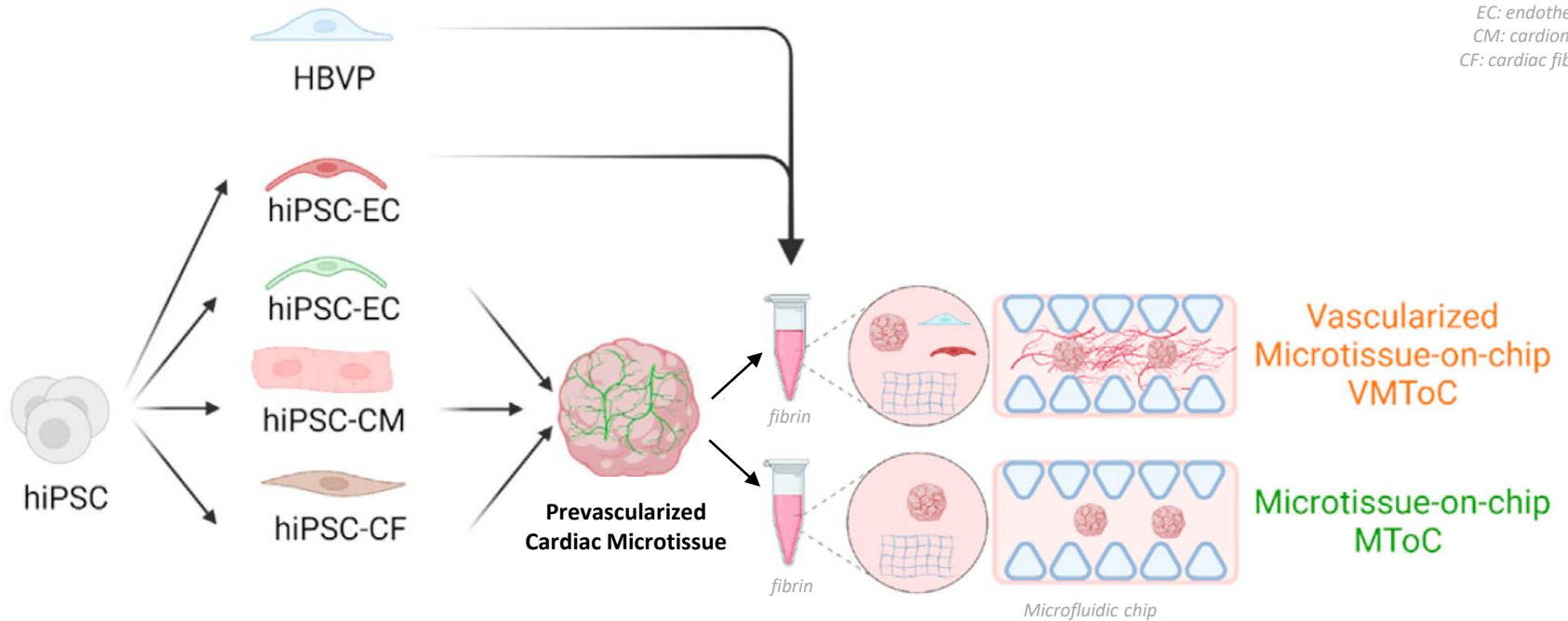
## **V. 3D Models**

# The different tissue models

|             |                                                                | Throughput                                                                               |                                                                                   |                                                                                              |                                                                                                       | Biological Relevance                                                                   |                                                                                     |
|-------------|----------------------------------------------------------------|------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|
|             |                                                                |         |  |             |                    |     |  |
|             |                                                                | Spheroids                                                                                | Microtissues                                                                      | Sheets                                                                                       | Strips                                                                                                | Rings                                                                                  | Chambers                                                                            |
| Advantages  | Readouts for <b>cell viability, apoptosis, motion analysis</b> | Readouts for <b>contraction force, cell viability, apoptosis</b>                         | Readouts for <b>optical mapping, motion analysis, Ca<sup>2+</sup> transients</b>  | Readouts for <b>contraction force, optical mapping, Ca<sup>2+</sup> transients</b>           | Readouts for <b>optical mapping, Ca<sup>2+</sup> transients, contraction force</b>                    | Readouts for <b>pressure-volume loops, optical mapping, Ca<sup>2+</sup> transients</b> |                                                                                     |
|             | Simple setup, automated seeding and analysis                   | Anisotropic organization, automated seeding and analysis, non-invasive force measurement | Larger surface area for optical mapping, amenable to electrical stimulation       | Non-invasive force measurement, anisotropic organization, amenable to electrical stimulation | Anisotropic organization, larger surface area for optical mapping, amenable to electrical stimulation | Clinically-relevant readouts, amenable to electrical stimulation                       |                                                                                     |
| Limitations | No direct force measurement, isotropic organization            | Requires specialized culture platforms, too small for optical mapping                    | No direct force measurement, anisotropic organization requires patterning         | Potential for drug absorption by silicone posts                                              | Direct force measurement requires force transducer, requires manual handling                          | Labor-intensive, requires specialized equipment                                        |                                                                                     |
|             |                                                                |                                                                                          |                                                                                   |                                                                                              |                                                                                                       |                                                                                        |                                                                                     |

# The spheroid model: an example

HBVP: Human Brain Vascular Pericytes  
EC: endothelial cells  
CM: cardiomyocytes  
CF: cardiac fibroblasts



## Experimental parameters

- Vascular network parameters
- Bead perfusion assay
- Sarcomere structure
- Contraction kinetics and amplitude
- Quantification of cytokines in the medium

## **VI. Pour se préparer à l'examen**

# Sujet d'examen

## Régénération cardiaque (durée conseillée : 45 minutes)

Basé sur *Early Regenerative Capacity in the Porcine Heart, Ye L. et al., Circulation. 2018;138:2798–2808*

*Il a été montré dans des modèles murins que le cœur des mammifères à la naissance possède une capacité de régénération efficace. Au contraire, il est connu que le cœur des mammifères adultes, même s'il possède une faible capacité d'auto-renouvellement, n'a pas la capacité de régénérer, notamment suite à un infarctus du myocarde.*

*Plusieurs équipes de recherche tentent de comprendre les mécanismes qui sous-tendent la régénération cardiaque. Ainsi une équipe a récemment publié un article qui étudie la capacité de régénération du cœur chez le porc nouveau-né (âgé de 2 jours).*

**Question 1.** En quelques lignes décrivez les arguments qui permettent de justifier l'intérêt de développer un tel modèle.

# Sujet d'examen

**Question 1.** En quelques lignes décrivez les arguments qui permettent de justifier l'intérêt de développer un tel modèle.

Structure argumentée

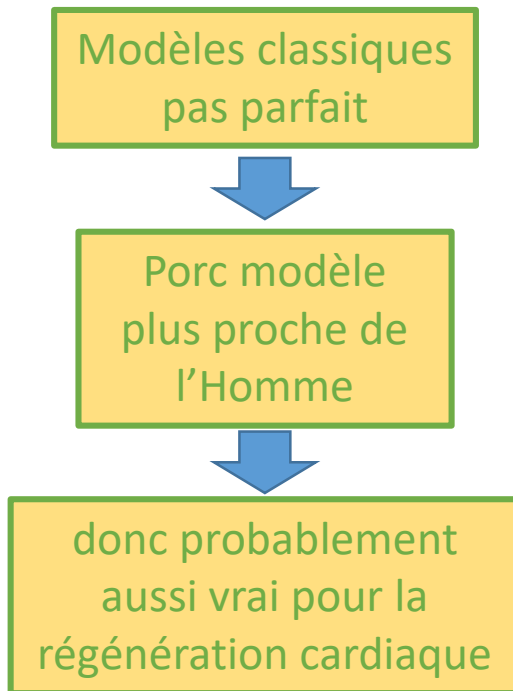
Précision de la réponse

# Sujet d'examen

**Question 1.** En quelques lignes décrivez les arguments qui permettent de justifier l'intérêt de développer un tel modèle.

Structure argumentée

Précision de la réponse

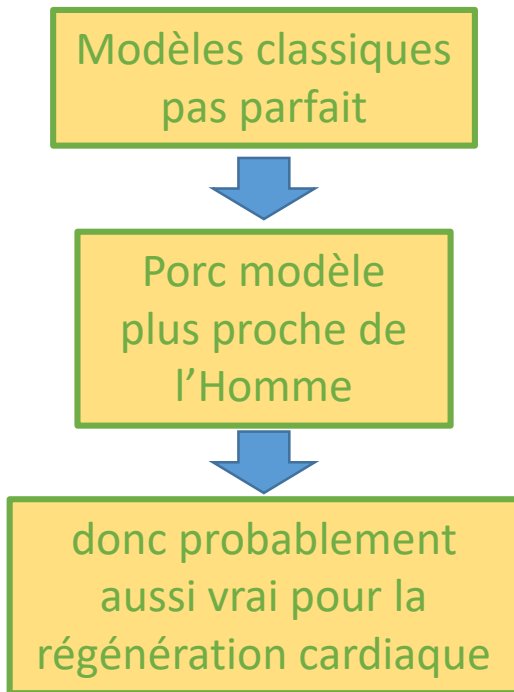


# Sujet d'examen

**Question 1.** En quelques lignes décrivez les arguments qui permettent de justifier l'intérêt de développer un tel modèle.

Structure argumentée

Précision de la réponse



Il est pertinent de développer un tel type de modèle car c'est un problème majeur de santé publique.

Les modèles classiques de régénération cardiaques sont le poisson-zèbre et la souris.

L'anatomie (structure de l'organe) est proche de celle de l'Homme contrairement aux modèles utilisés couramment pour étudier la régénération cardiaque. (ventricules, oreillettes, 2 circuits)

Cependant, la fonction cardiaque d'un porc est proche de celle de l'Homme contrairement aux modèles utilisés couramment pour étudier la régénération cardiaque (rythme, pressions).

Cette proximité suggère que le porc pourrait avoir des capacités de régénération identique à l'Homme.

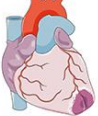
# Sujet d'examen

**Question 2.** Il s'agit ici d'un modèle chirurgical par ligature de l'artère coronaire gauche. Citer au moins un autre type de modèle chirurgical permettant de créer une lésion cardiaque permettant d'étudier la régénération cardiaque.

# Sujet d'examen

**Question 2.** Il s'agit ici d'un modèle chirurgical par ligature de l'artère coronaire gauche. Citer au moins un autre type de modèle chirurgical permettant de créer une lésion cardiaque permettant d'étudier la régénération cardiaque.

| Surgical Models                                                                                                                                                                 | Endovascular Models                                                                                                                                                                                                 |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Pro:</b> <ul style="list-style-type: none"> <li>- Very precise</li> <li>- More variation between procedures</li> <li>- Implantation of extra monitoring equipment</li> </ul> | <b>Pro:</b> <ul style="list-style-type: none"> <li>- Lower labor intensity</li> <li>- Faster (allows for higher volume)</li> <li>- Lower peri- / post-OP complications</li> <li>- Less strain on animals</li> </ul> |
| <b>Con:</b> <ul style="list-style-type: none"> <li>- High labor intensity</li> <li>- Slow</li> <li>- More post-OP complications</li> <li>- More strain on animals</li> </ul>    | <b>Con:</b> <ul style="list-style-type: none"> <li>- Limited devices / methodologies</li> <li>- Imaging required</li> <li>- No implantation of extra equipment</li> </ul>                                           |

| Ischemia                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Acute:</b> - Artery ligation ( $\pm$ reperfusion)<br><b>Chronic:</b> Ameroid constrictor                                                                 |
|                                                                            |
| <b>Acute:</b> - Balloon occlusion ( $\pm$ reperfusion)<br><b>Chronic:</b> Repeat microembolization<br>- Bottleneck/Hourglass stent<br>- Copper-plated stent |

| Pressure Overload                                                                                  |
|----------------------------------------------------------------------------------------------------|
| <b>Acute:</b> - Aortic banding / cuffing<br>- Banding of A. renalis<br>- s.c. DOCA pellets (+diet) |
|                   |
| <b>Chronic:</b> - Stenting of young aorta desc.<br>- Stenting of A. renalis                        |

| Volume Overload                                                                    |
|------------------------------------------------------------------------------------|
| <b>Acute:</b> - Surgical mitral regurgitation<br>- Arteriovenous fistula (AVF)     |
|  |
| <b>Chronic:</b> - Cutting mitral chordae                                           |

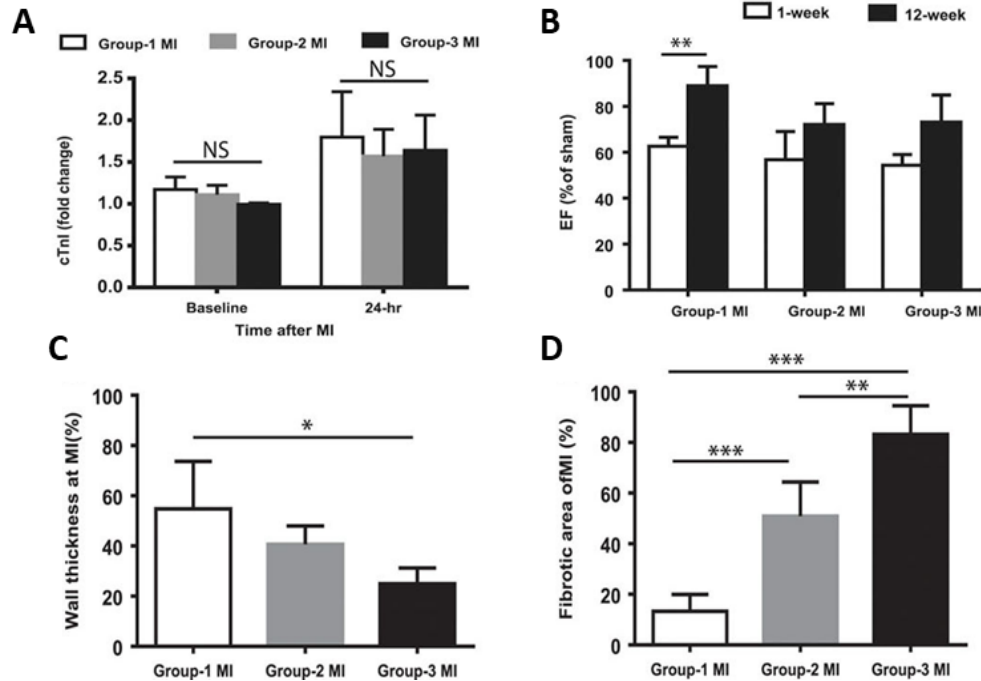
| Cytotoxic                                                                           |
|-------------------------------------------------------------------------------------|
| <b>Acute:</b> - Doxorubicin (i.v.) + AVF                                            |
|  |
| <b>Chronic:</b> - Doxorubicin (intracoronary)                                       |

| Pacing                                                                               |
|--------------------------------------------------------------------------------------|
| <b>Acute:</b> - Epicardial pacing leads<br>- Single vs. Multiple leads               |
|   |
| <b>Chronic:</b> - Atrial vs. Ventricular pacing leads<br>- Single vs. Multiple leads |

| Type              | Method                                                     | Advantages                                                                                 | Disadvantages                                                                             | Literature                                                 |
|-------------------|------------------------------------------------------------|--------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------|
| Acute ischemia    | Embolization                                               | Low-complexity, cheap                                                                      | Less reproducible, less precise, no reperfusion possible                                  | Canine (10)<br>Porcine (11–13)<br>Ovine (14, 15)           |
|                   | LAD ligation                                               | Reproducible, possibility of reperfusion, extra-coronary device implantation possible      | Open-chest, lethal arrhythmias, surgical complications, labor intensive,                  | Canine (16)<br>Porcine (17, 18)<br>Ovine (19, 20)          |
|                   | Balloon occlusion                                          | Reproducible, possibility of reperfusion, closed-chest, fewer complications                | Less precise than surgical ligation                                                       | Canine (21)<br>Porcine (22–27)<br>Ovine (28)               |
| Chronic ischemia  | Repeated microembolization                                 | Good model of microvascular disease                                                        | Repeat interventions, labor intensive                                                     | Canine (29, 30)<br>Ovine (31, 32)<br>Bovine (33)           |
|                   | Ameroid constrictor, bottleneck stent, copper-plated stent | Good model of single vessel disease                                                        | Variable degree of stenosis and LVEF reduction                                            | Canine (34)<br>Porcine (35–42)<br>Ovine (43)               |
| Pressure overload | Surgical aortic banding                                    | Reproducible, minimal surgical procedure                                                   | Abrupt increase in pressure                                                               | Canine (44)<br>Porcine (45, 46)<br>Ovine (47)              |
|                   | Stenting of descending aorta                               | Good model of aortic stenosis, gradual pressure overload, very similar to human conditions | Long time is necessary to develop left ventricular hypertrophy and dilation               | Porcine (48)                                               |
|                   | Renal artery stenosis                                      | Good model of severe systemic hypertension, low-complexity, single intervention            | Hormonal dysregulation, difficult to control                                              | Canine (49, 50)<br>Porcine (51)<br>Ovine (52)              |
|                   | DOCA pellet s.c. implantation with high salt diet          | Simple non-invasive method                                                                 | Mild systemic hypertension, uncertain effect on left ventricular function                 | Porcine (53)                                               |
| Volume overload   | Mitral regurgitation                                       | Good model of volume overload, reproducible                                                | Technically difficult                                                                     | Canine (54–57)<br>Porcine (58)<br>Ovine (59)               |
|                   | Arteriovenous fistula                                      |                                                                                            | Low clinical relevance, not well-established in large animals                             | Goat (60)<br>Ovine (61)                                    |
| Cytotoxic         | Anthracyclines                                             | Good model of doxorubicin-induced heart failure, simple non-invasive method                | Narrow clinical relevance, toxic substances, often only in combination with other methods | Canine (62, 63)<br>Porcine (64)<br>Ovine (65)<br>Goat (60) |
| Pacing            | Rapid pacing                                               | Good model of tachycardic remodeling                                                       | Reversible HF, qualitatively different remodeling                                         | Canine (66, 67)<br>Porcine (68–72)<br>Ovine (73)           |

Summarizes the different kinds of HF/EF models, their main uses and their advantages and disadvantages. The selected papers are meant to serve as illustrative examples of the various models, they do not represent a comprehensive list of all studies employing the respective model. LAD, Left anterior descending; LVEF, Left ventricular ejection fraction; DOCA, deoxycorticosterone acetate; sc, subcutaneous; HF, Heart Failure.

# Sujet d'examen



**Figure 1** : Trois groupes de cochons ont subi un infarctus chirurgical (ligature de l'artère coronaire gauche). Les cochons du groupe 1 sont âgés de 2 jours (après la naissance), ceux du groupe 2 sont âgés de 3 jours et ceux du groupe 3 sont âgés de 14 jours. A, quantification de la troponine I cardiaque (cTnI, protéine spécifiquement exprimée dans les cardiomyocytes) dans le sang juste avant la chirurgie (baseline) où 24h après la chirurgie. B, Fraction d'éjection cardiaque des 3 groupes de cochons 1 semaine et 12 semaines après la chirurgie. C, analyse semi-quantitative de l'épaisseur de la paroi du ventricule gauche 12 semaine après la chirurgie. D, analyse semi-quantitative de l'aire de la zone fibreuse du ventricule gauche 12 semaines après chirurgie.

**Question 3.** Selon vous, par quelle technique les résultats présentés dans ces panneaux sont-ils obtenus et comment sont-ils normalisés ?

# Sujet d'examen

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Panneau A. Quantification de la protéine cTnI sanguine par ELISA.

Panneaux B et C. Par Imagerie par résonance magnétique (IRM).

Panneau D. Après prélèvement, coupe et coloration histologique.

La normalisation est faite par rapport à la valeur moyenne obtenue pour le groupe Sham.